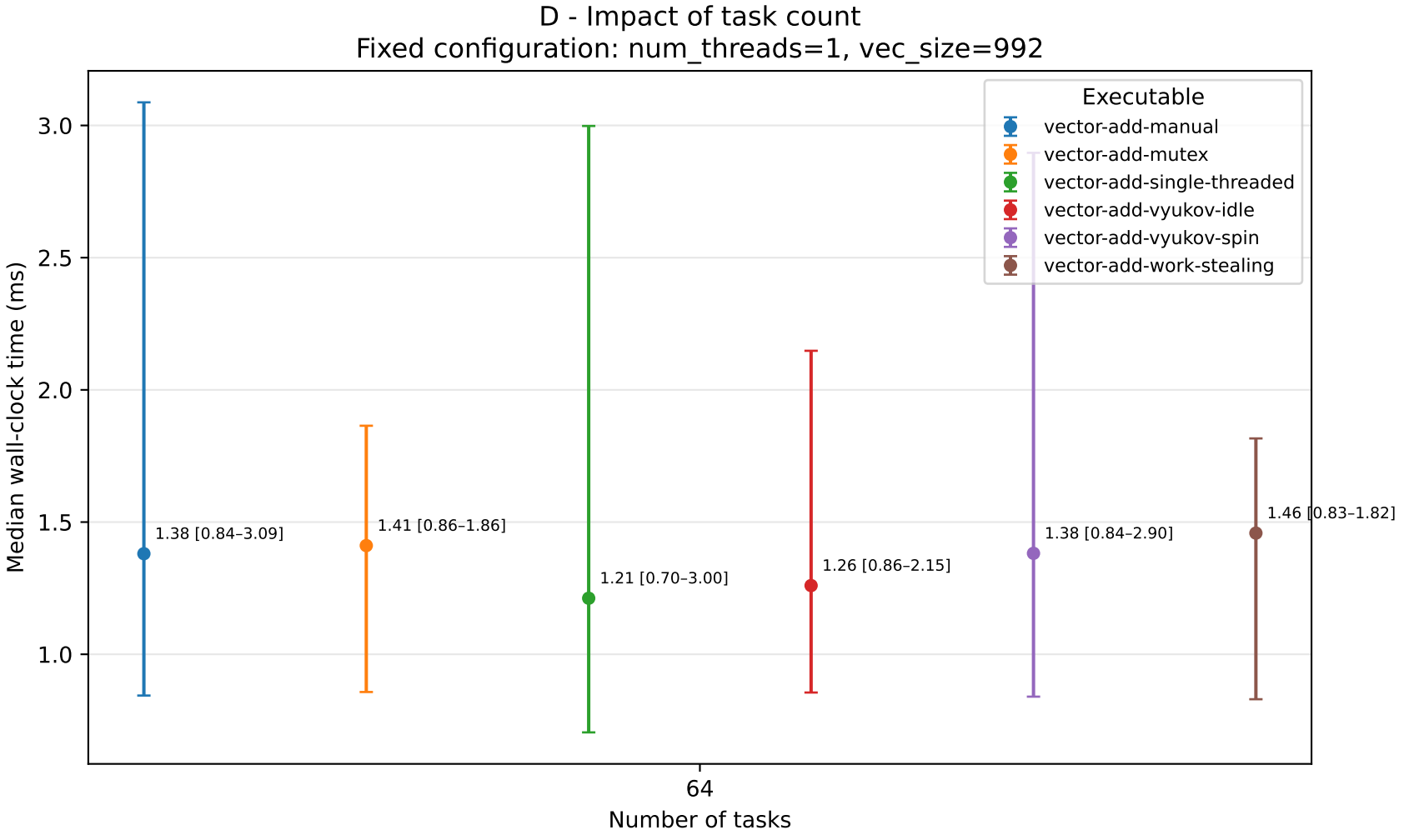
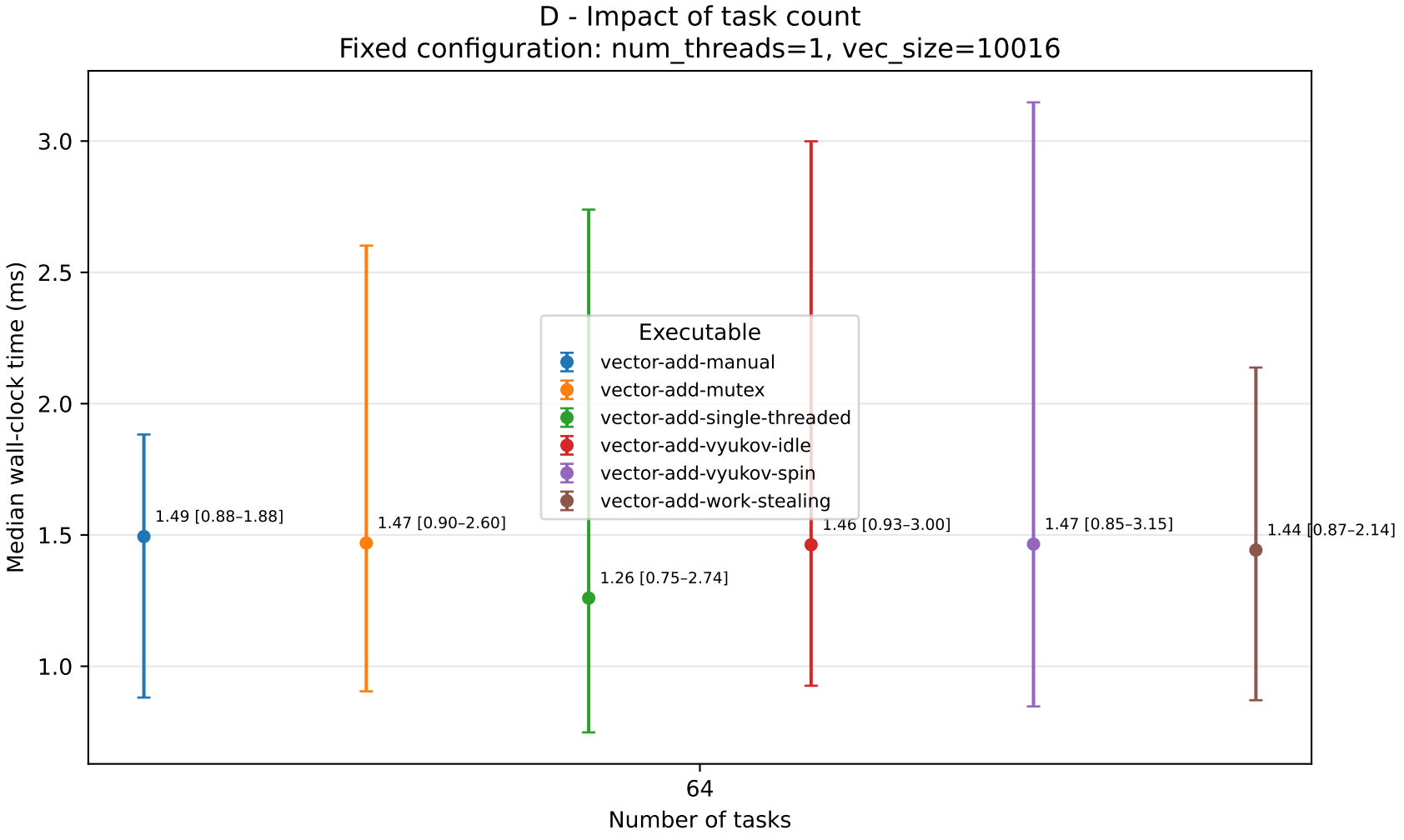
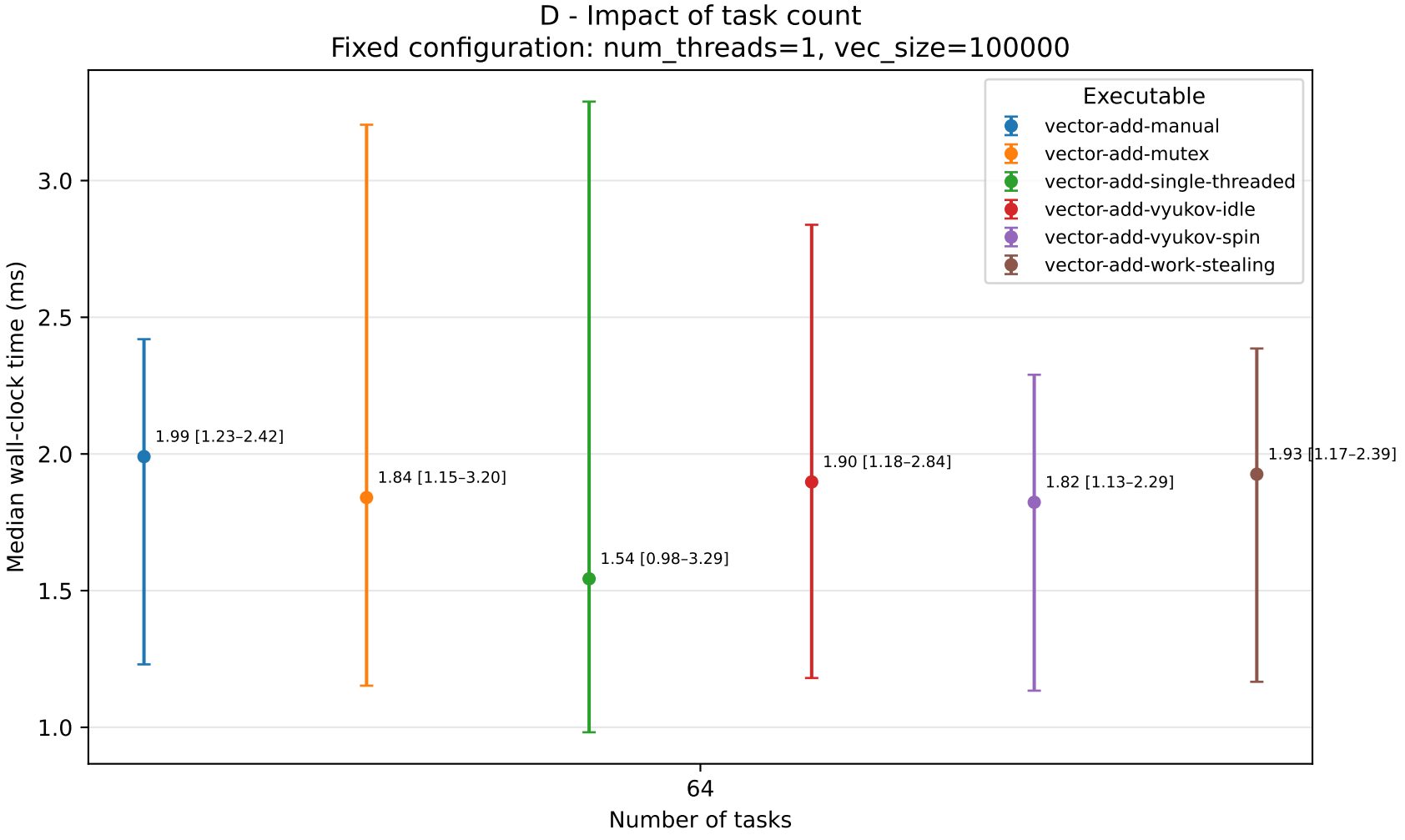
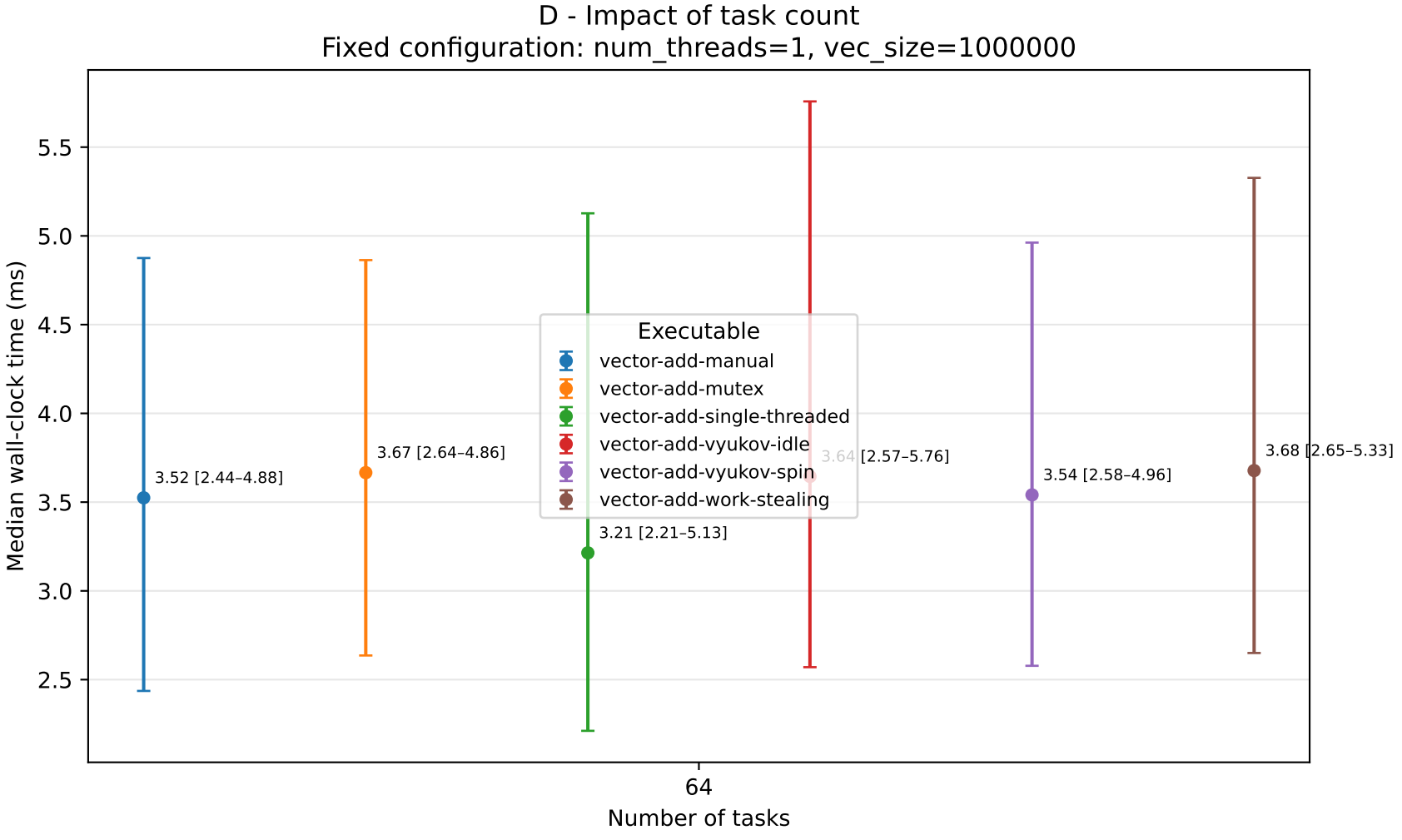
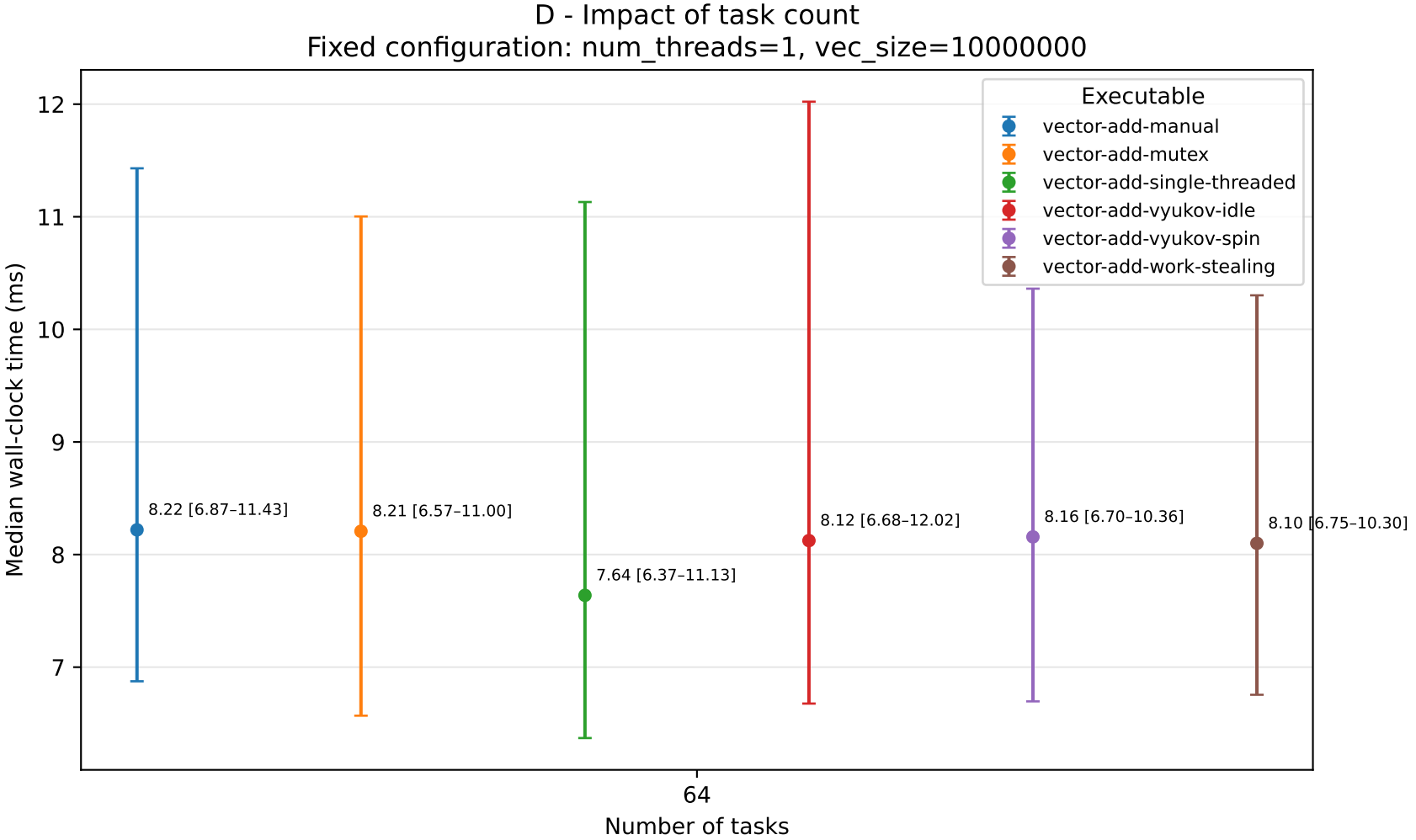
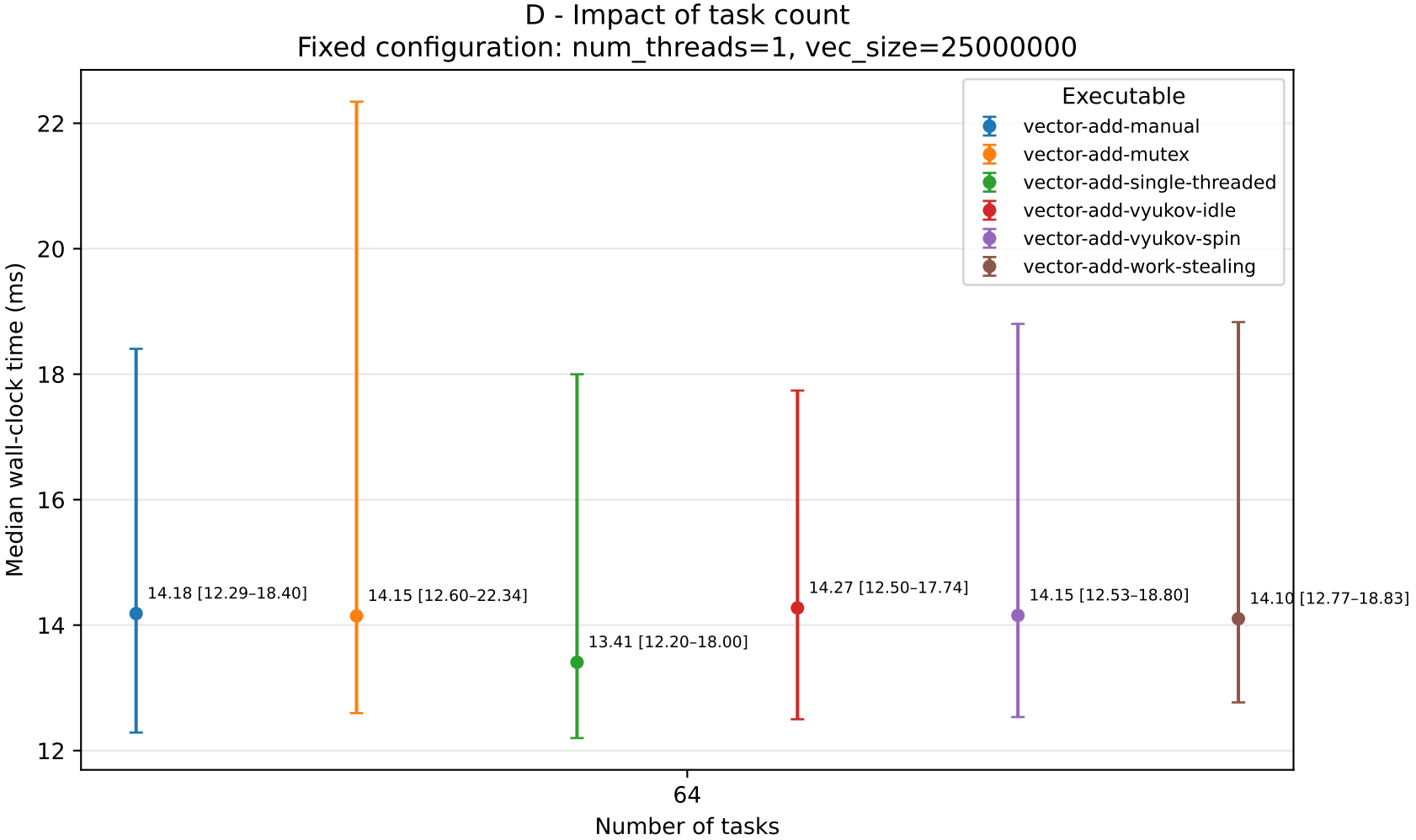

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=992
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.38 [0.84–3.09]
1.41 [0.86–1.86]
1.21 [0.70–3.00]
1.26 [0.86–2.15]
1.38 [0.84–2.90]
1.46 [0.83–1.82]
64
Number of tasks

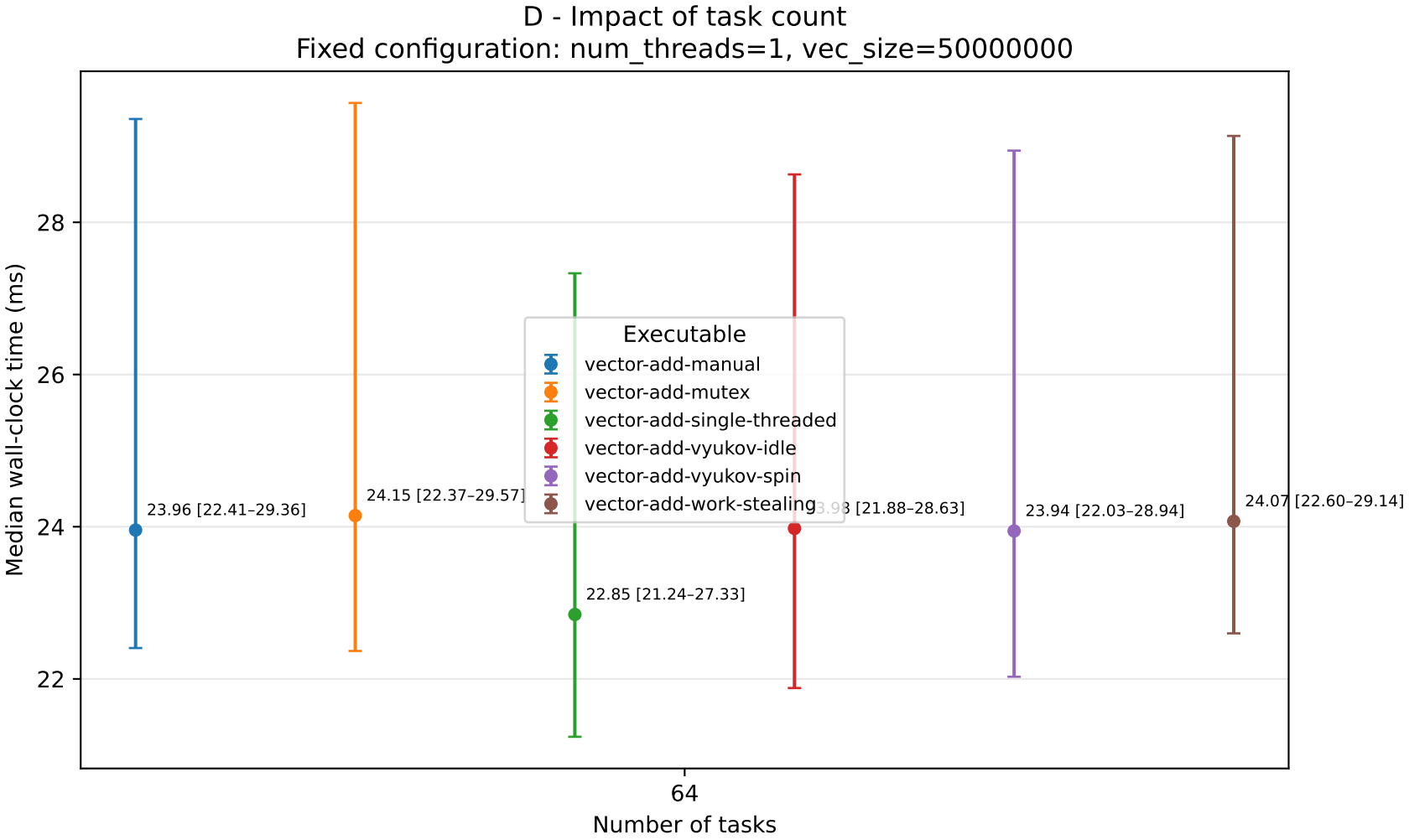

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=10016
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
1.49 [0.88–1.88]
1.47 [0.90–2.60]
1.26 [0.75–2.74]
1.46 [0.93–3.00]
1.47 [0.85–3.15]
1.44 [0.87–2.14]
64
Number of tasks

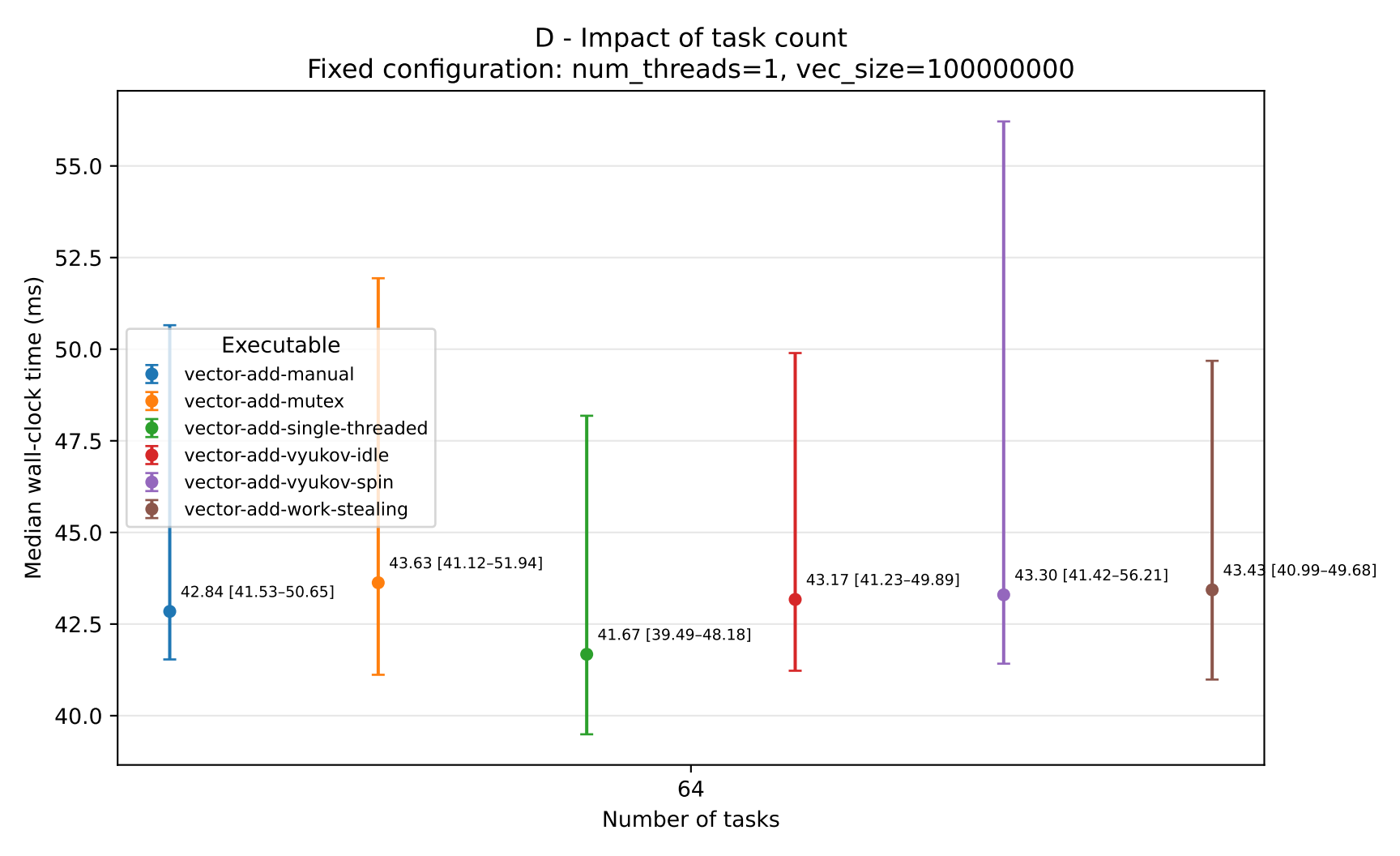

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=100000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.99 [1.23–2.42]
1.84 [1.15–3.20]
1.54 [0.98–3.29]
1.90 [1.18–2.84]
1.82 [1.13–2.29]
1.93 [1.17–2.39]
64
Number of tasks

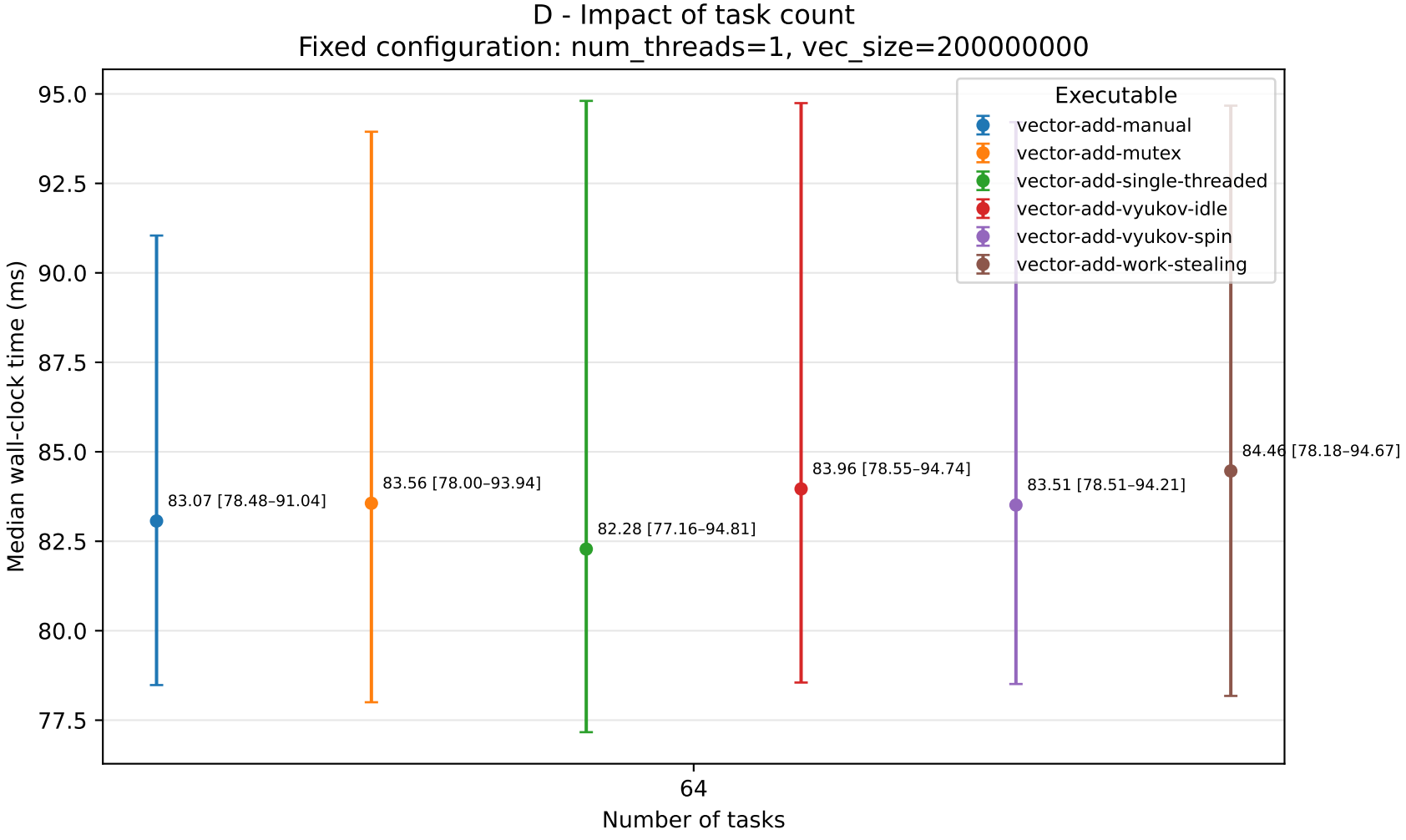

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=1000000
Median wall-clock time (ms)
5.5
5.0
4.5
4.0
3.5
3.0
2.5
3.52 [2.44–4.88]
3.67 [2.64–4.86]
3.21 [2.21–5.13]
3.64 [2.57–5.76]
3.54 [2.58–4.96]
3.68 [2.65–5.33]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks

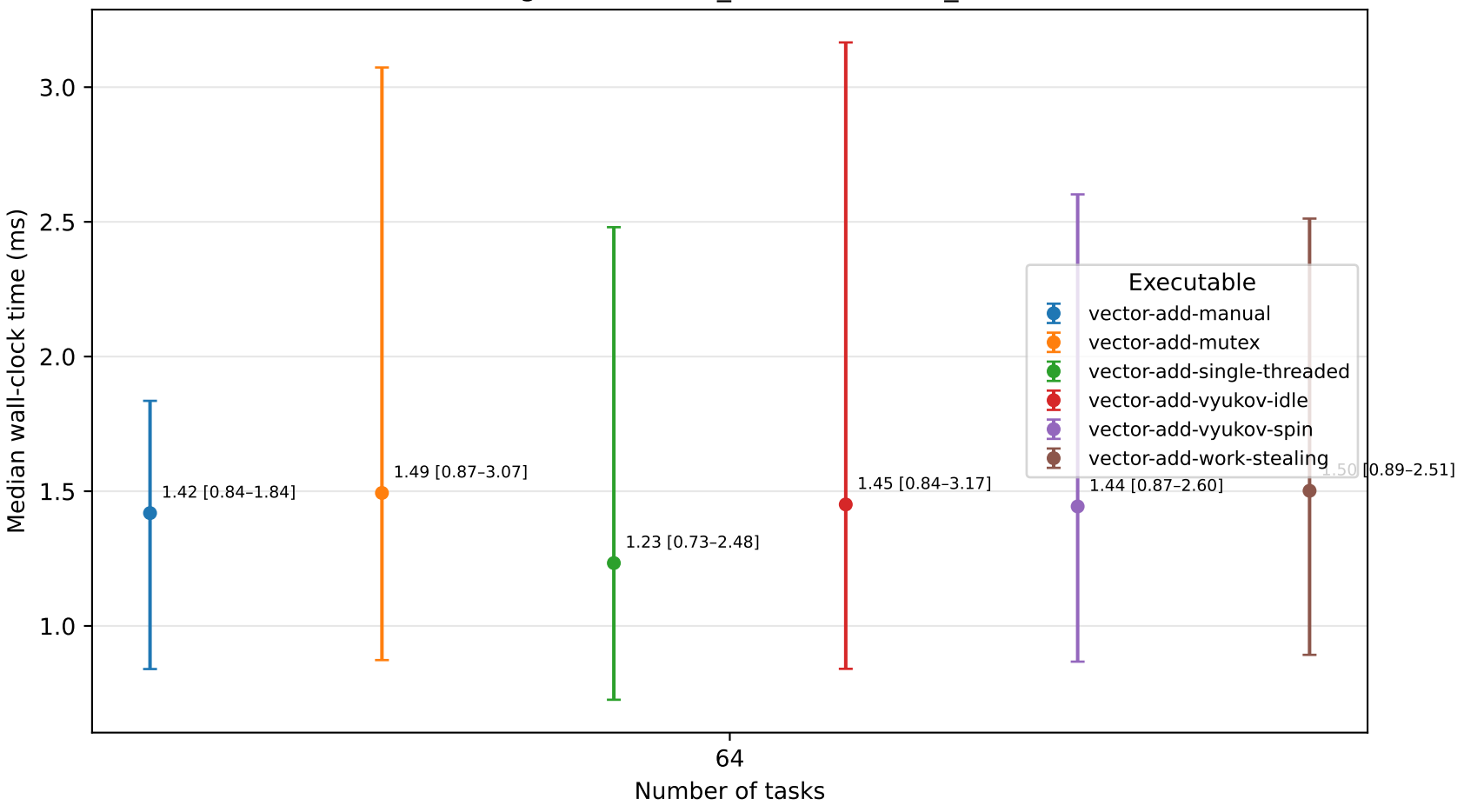

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=10000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
12
11
10
9
8
7
8.22 [6.87–11.43]
8.21 [6.57–11.00]
7.64 [6.37–11.13]
8.12 [6.68–12.02]
8.16 [6.70–10.36]
8.10 [6.75–10.30]
64
Number of tasks

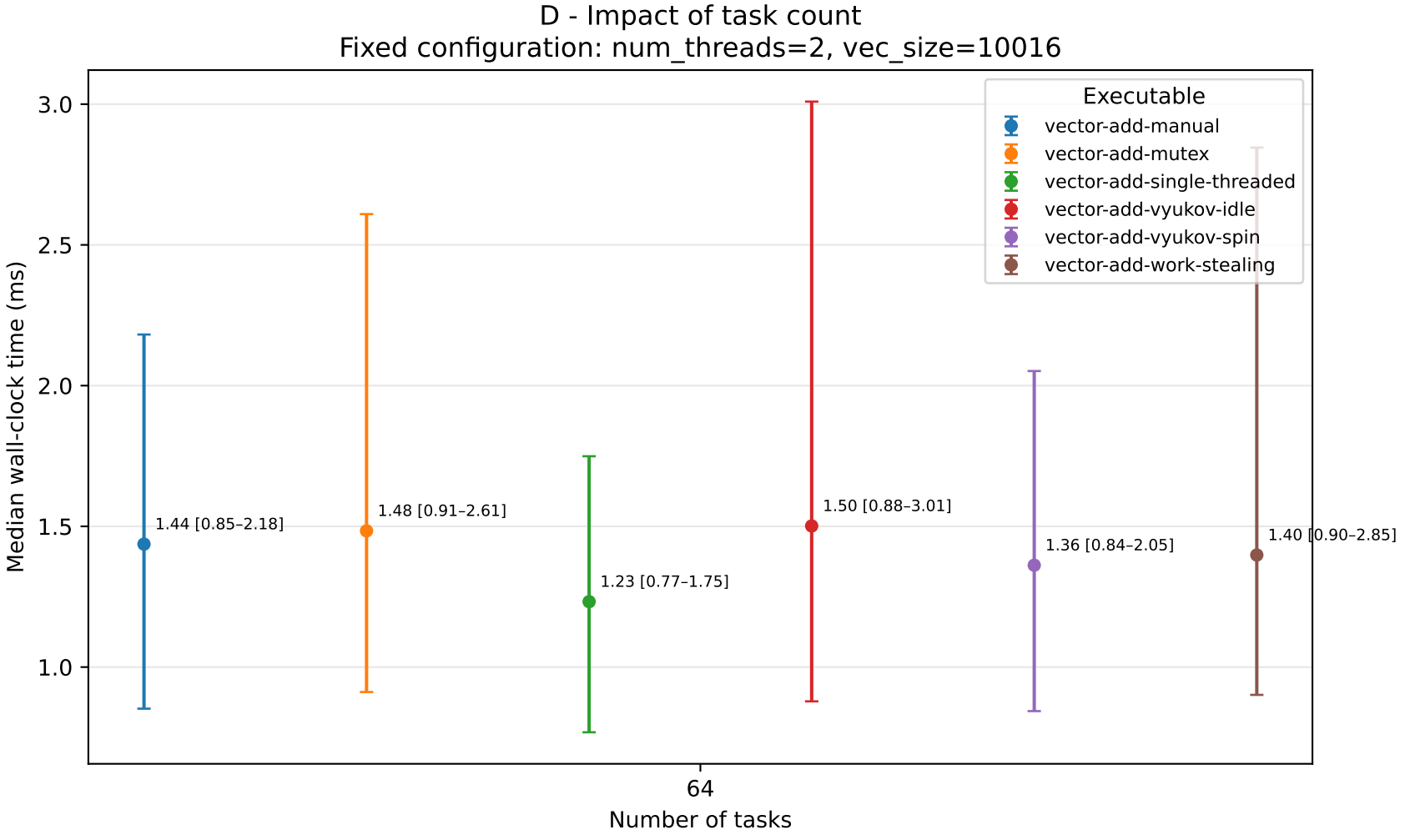

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=25000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
22
20
18
16
14
12
14.18 [12.29–18.40]
14.15 [12.60–22.34]
13.41 [12.20–18.00]
14.27 [12.50–17.74]
14.15 [12.53–18.80]
14.10 [12.77–18.83]
64
Number of tasks

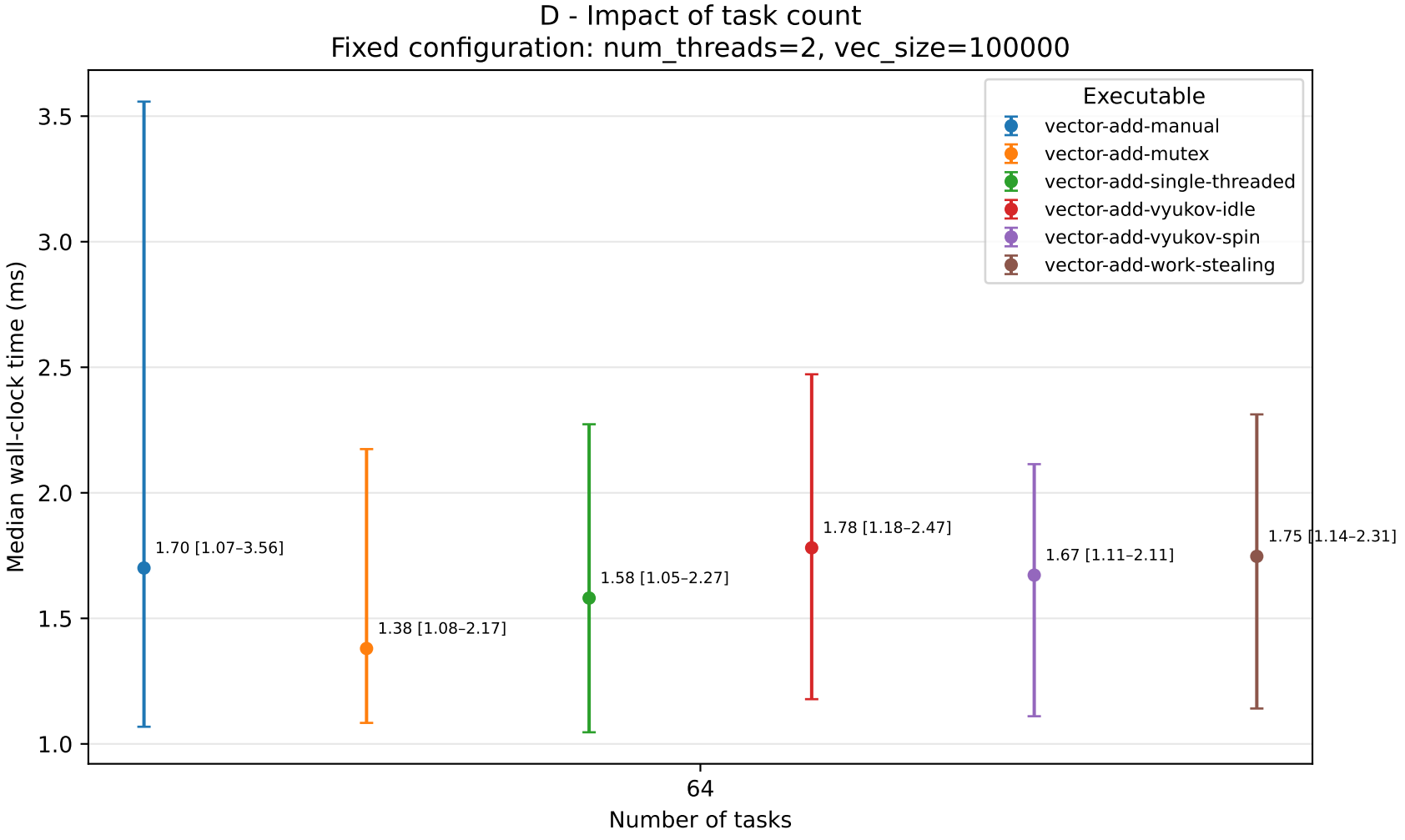

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=50000000
Median wall-clock time (ms)
28
26
24
22
64
Number of tasks
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
23.96 [22.41–29.36]
24.15 [22.37–29.57]
22.85 [21.24–27.33]
23.94 [22.03–28.94]
24.07 [22.60–29.14]

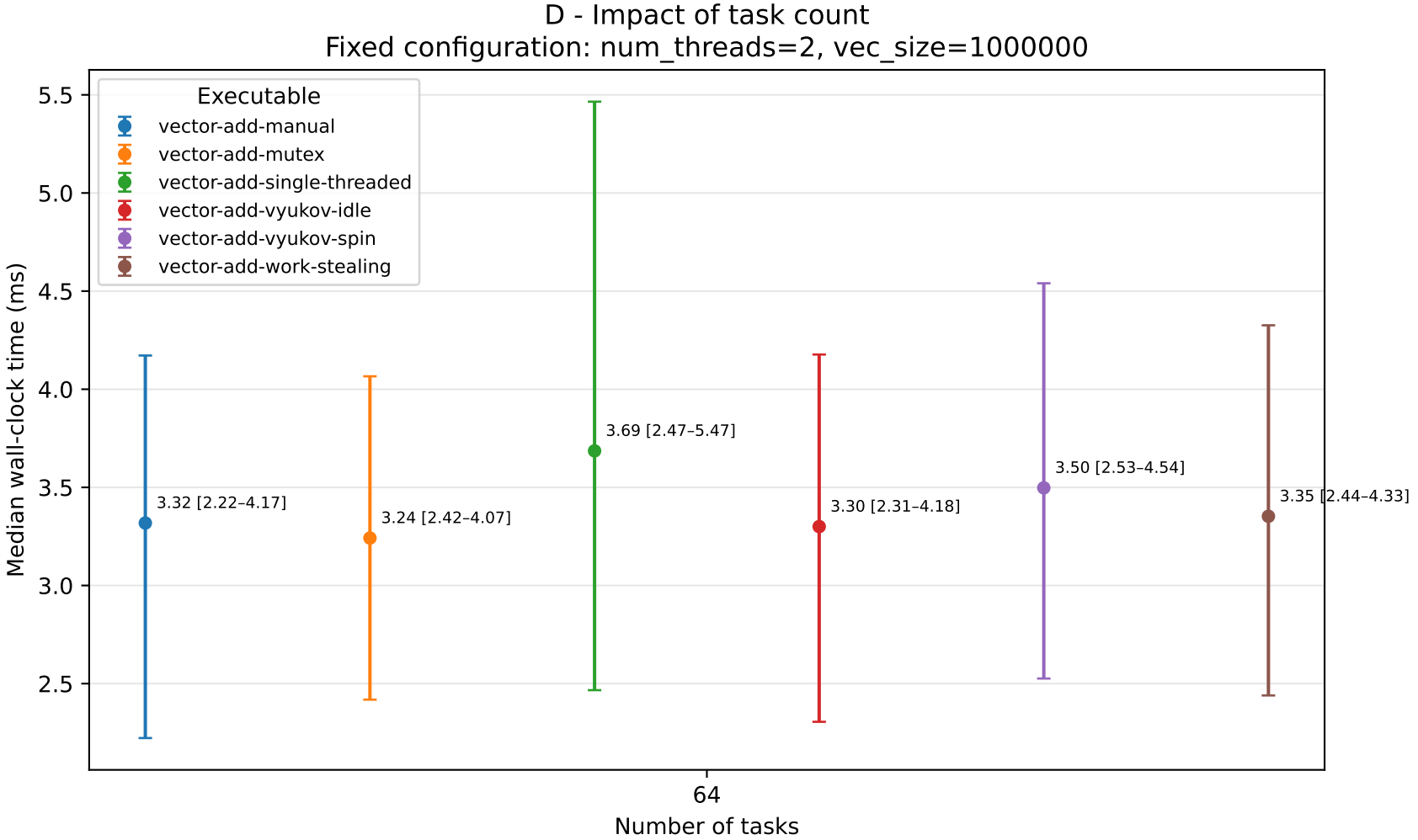

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=100000000
Median wall-clock time (ms)
55.0
52.5
50.0
47.5
45.0
42.5
40.0
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
42.84 [41.53–50.65]
43.63 [41.12–51.94]
41.67 [39.49–48.18]
43.17 [41.23–49.89]
43.30 [41.42–56.21]
43.43 [40.99–49.68]
64
Number of tasks

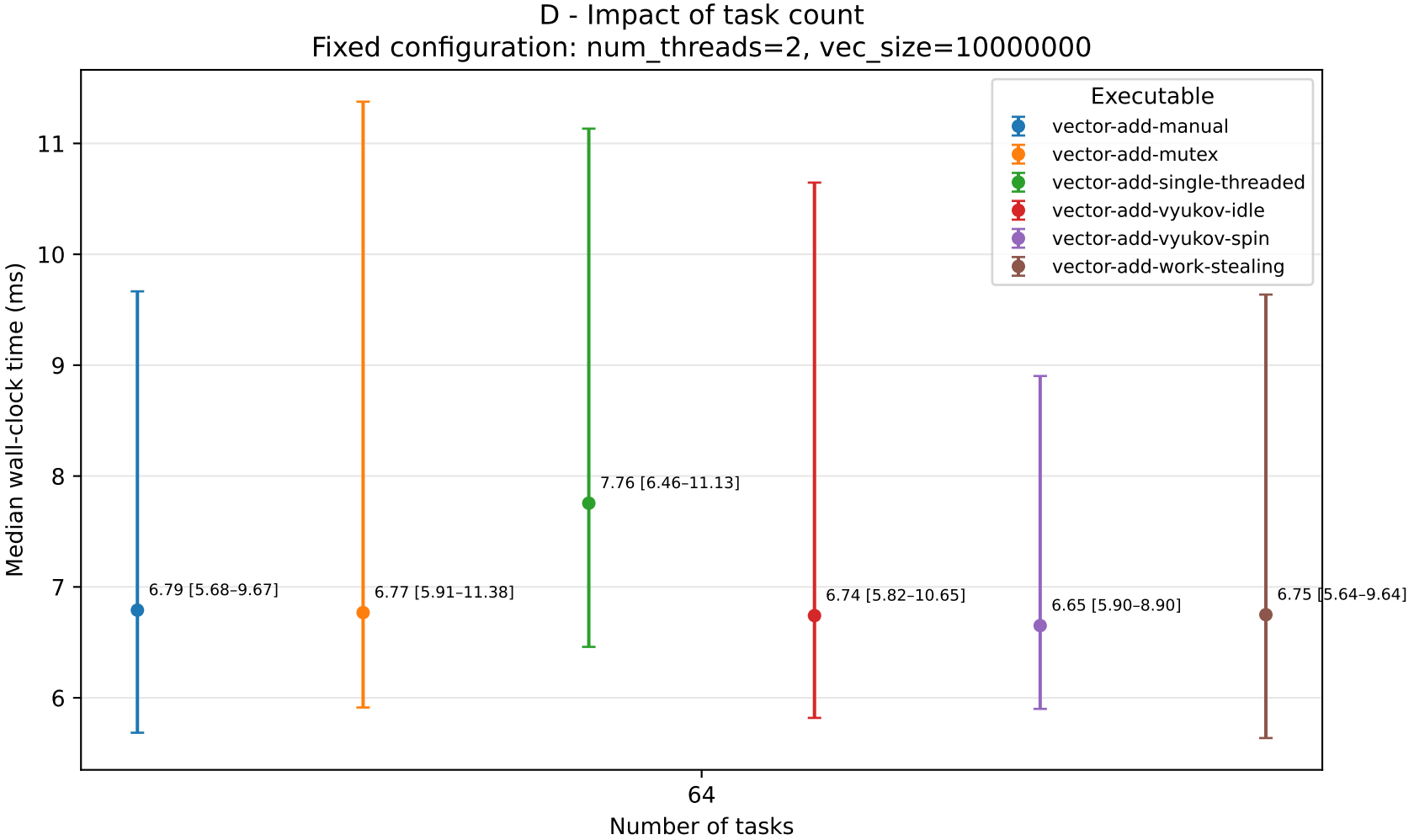

D - Impact of task count
Fixed configuration: num_threads=1, vec_size=200000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
95.0
92.5
90.0
87.5
85.0
82.5
80.0
77.5
83.07 [78.48–91.04]
83.56 [78.00–93.94]
82.28 [77.16–94.81]
83.96 [78.55–94.74]
83.51 [78.51–94.21]
84.46 [78.18–94.67]
64
Number of tasks

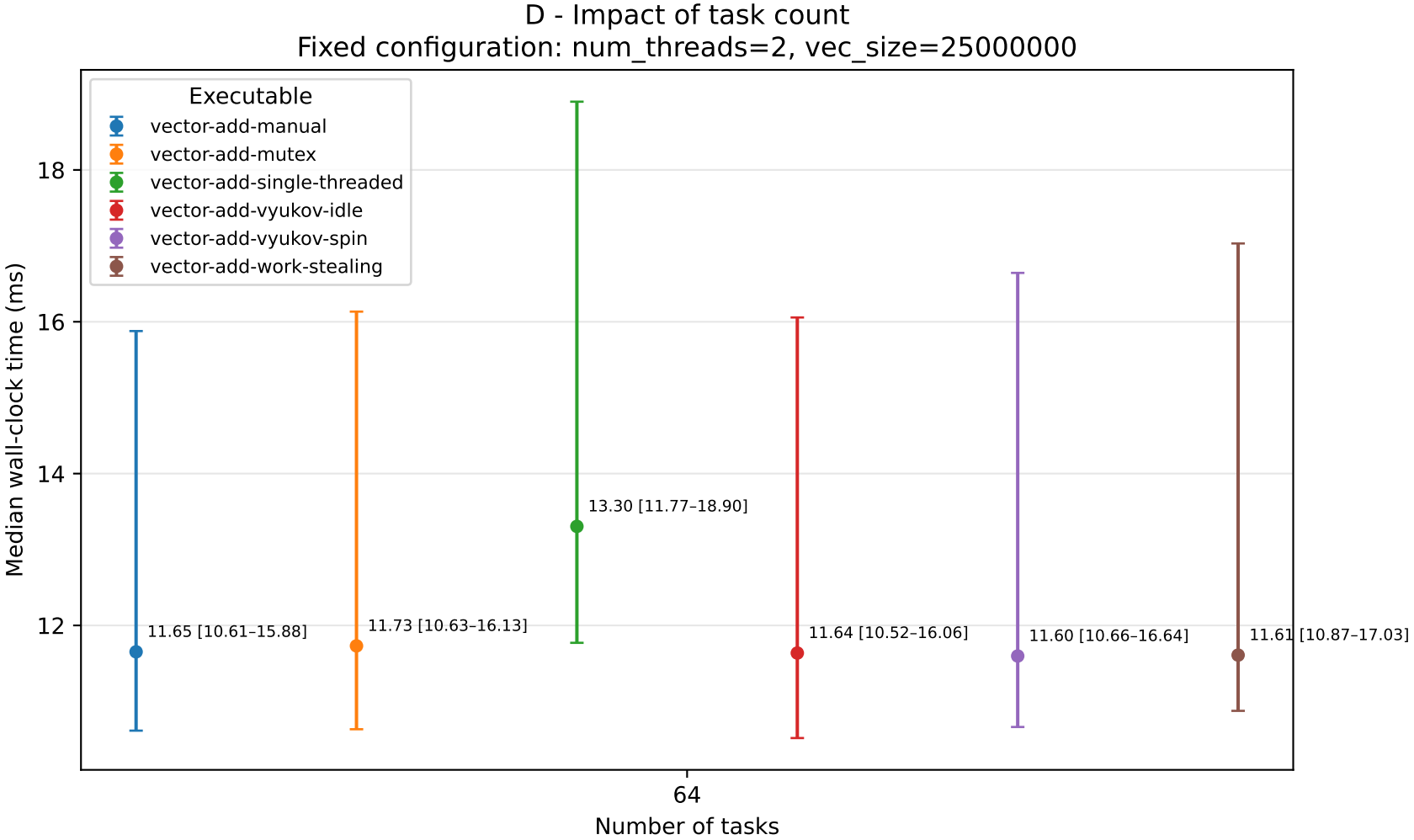

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=992
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.42 [0.84–1.84]
1.49 [0.87–3.07]
1.23 [0.73–2.48]
1.45 [0.84–3.17]
1.44 [0.87–2.60]
1.50 [0.89–2.51]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks

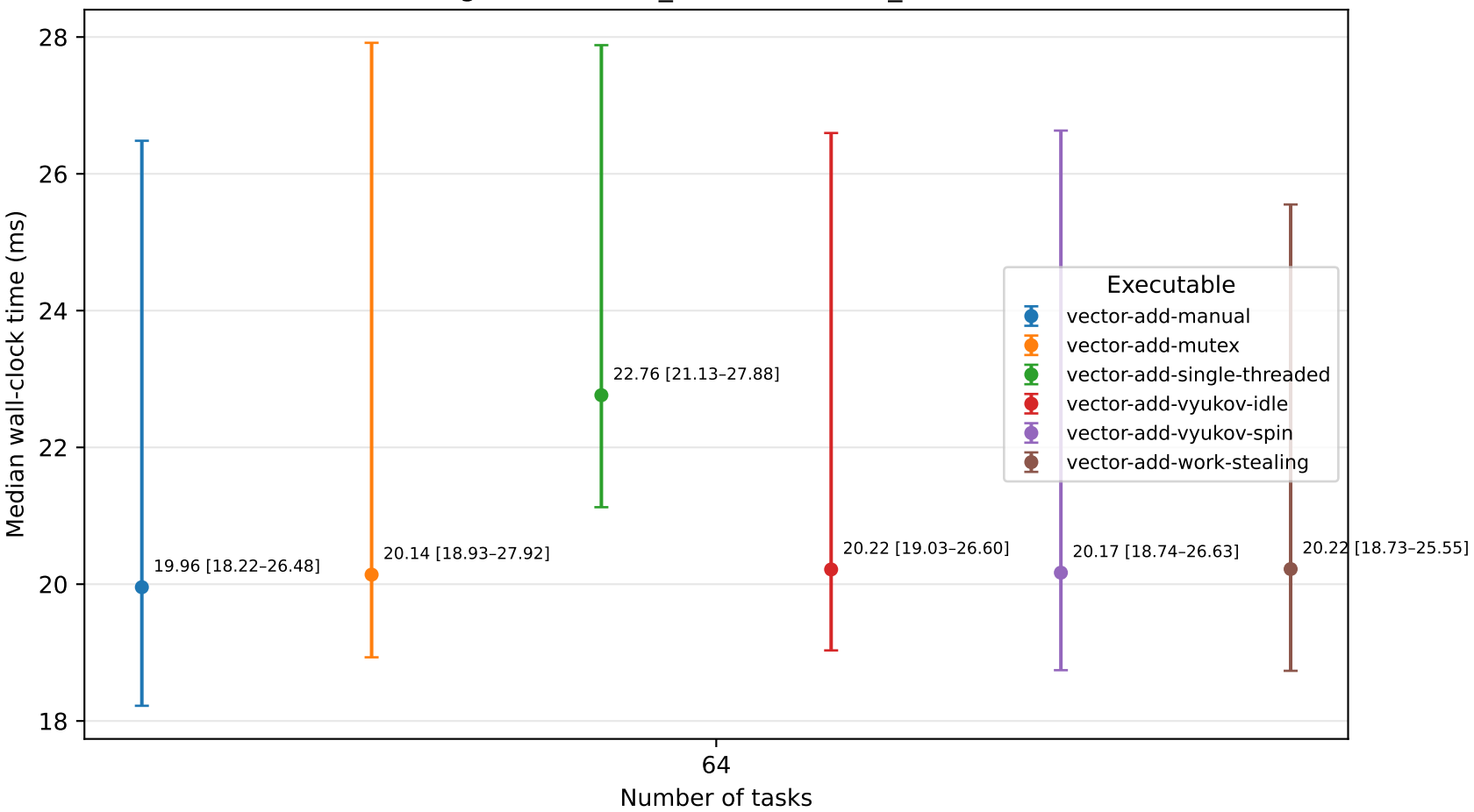

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=10016
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.44 [0.85–2.18]
1.48 [0.91–2.61]
1.23 [0.77–1.75]
1.50 [0.88–3.01]
1.36 [0.84–2.05]
1.40 [0.90–2.85]
64
Number of tasks

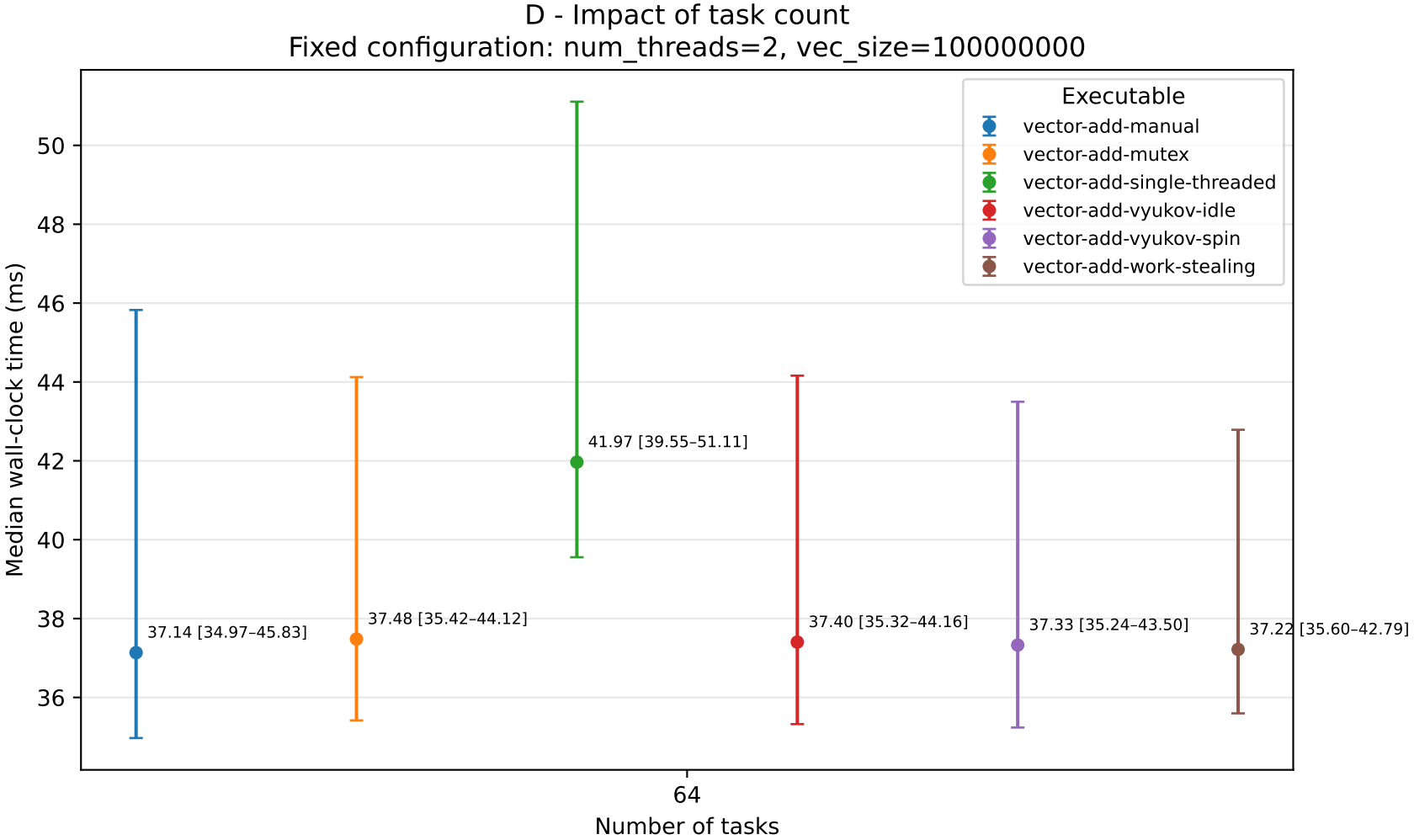

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=100000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
3.5
3.0
2.5
2.0
1.5
1.0
1.70 [1.07–3.56]
1.38 [1.08–2.17]
1.58 [1.05–2.27]
1.78 [1.18–2.47]
1.67 [1.11–2.11]
1.75 [1.14–2.31]
64
Number of tasks

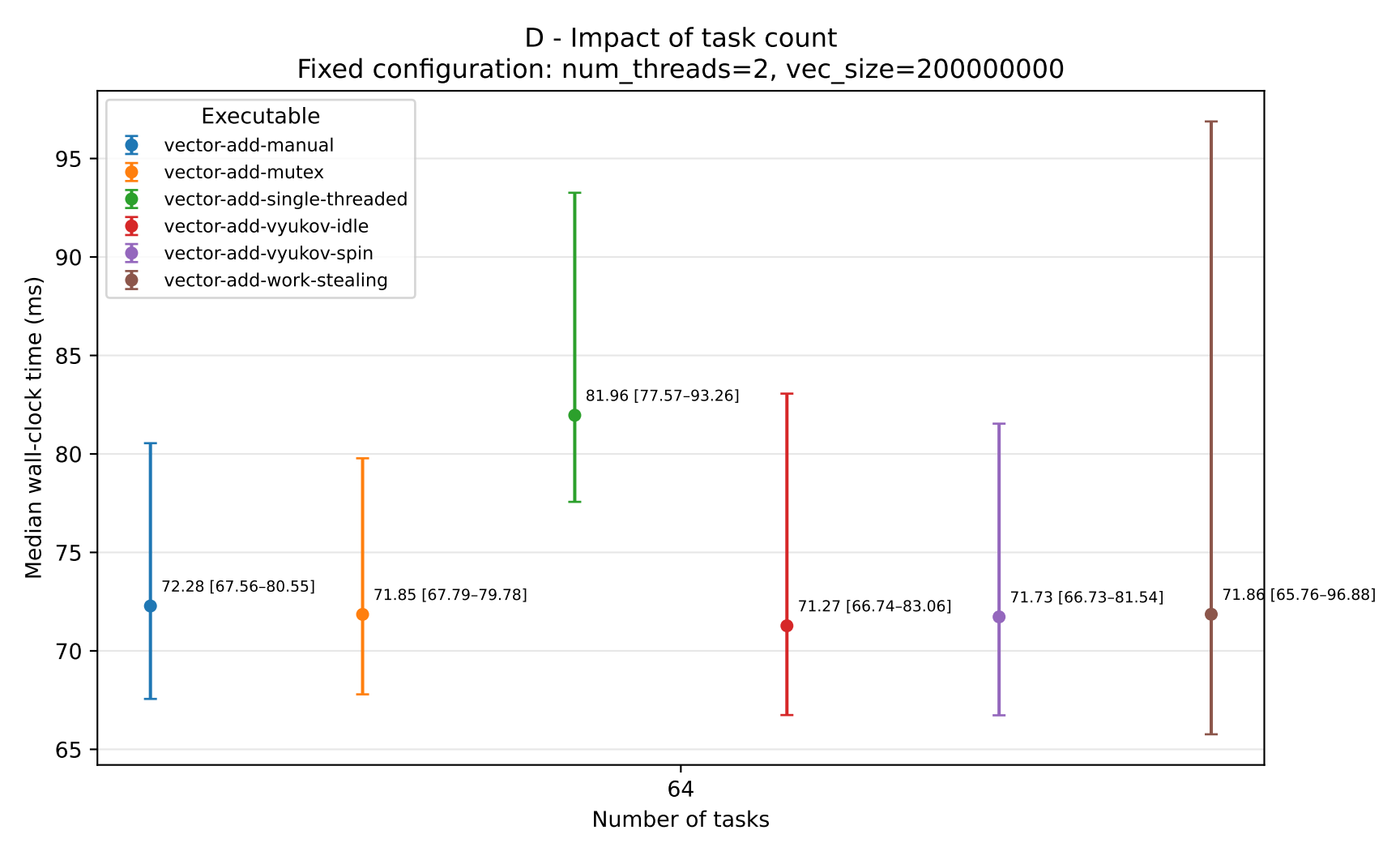

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=1000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
5.5
5.0
4.5
4.0
3.5
3.0
2.5
3.32 [2.22–4.17]
3.24 [2.42–4.07]
3.69 [2.47–5.47]
3.30 [2.31–4.18]
3.50 [2.53–4.54]
3.35 [2.44–4.33]
64
Number of tasks

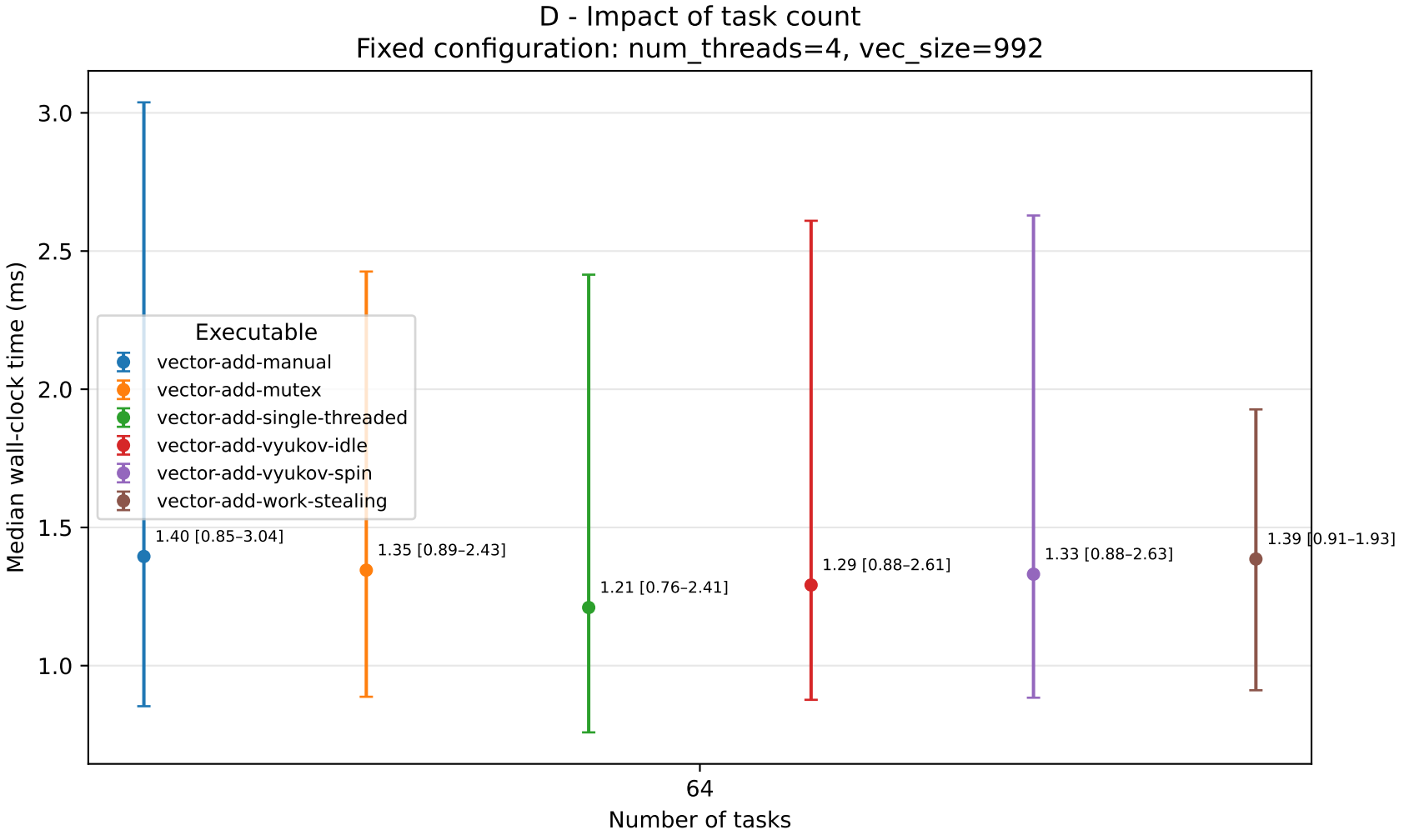

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=10000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
11
10
9
8
7
6
6.79 [5.68–9.67]
6.77 [5.91–11.38]
7.76 [6.46–11.13]
6.74 [5.82–10.65]
6.65 [5.90–8.90]
6.75 [5.64–9.64]
64
Number of tasks

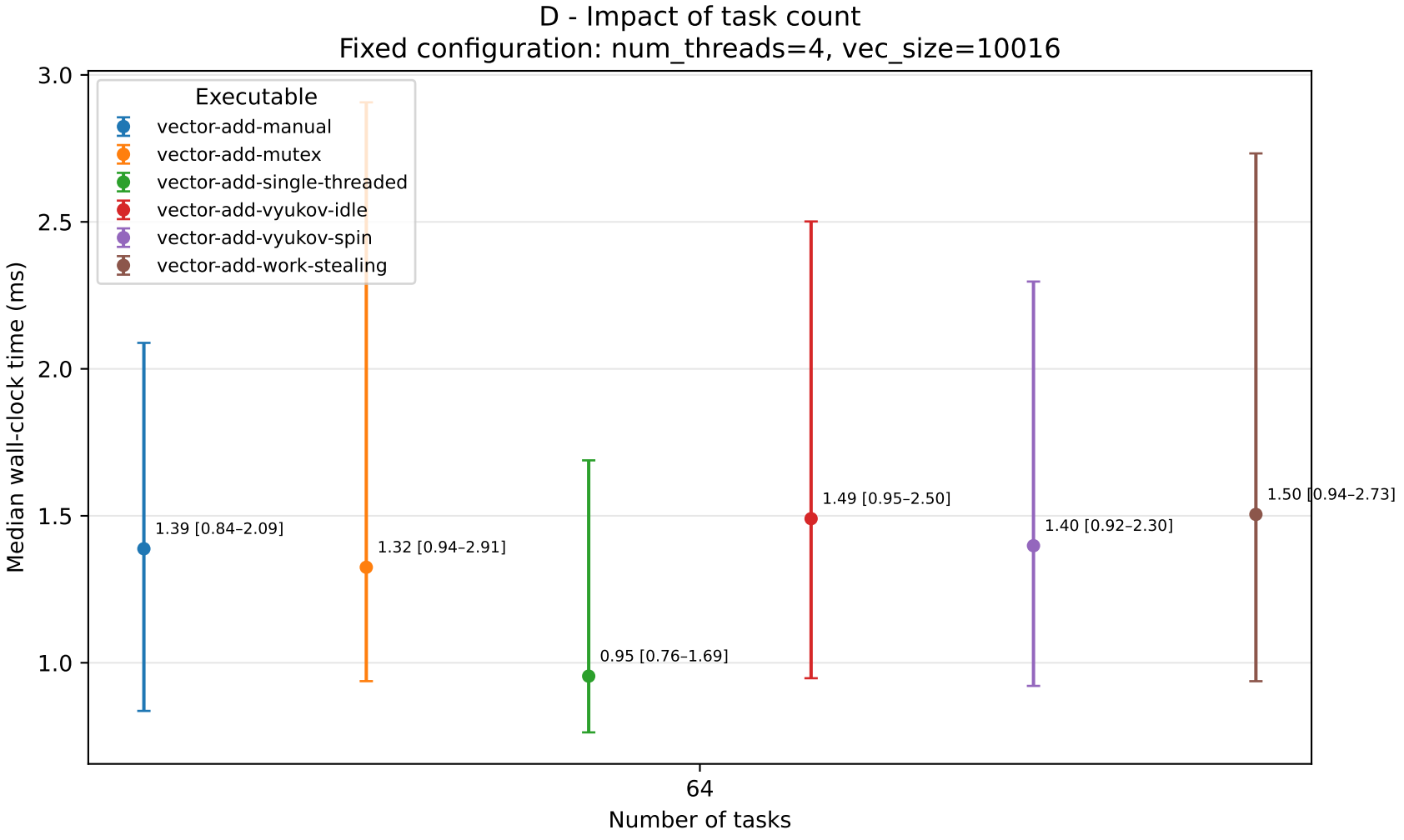

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=25000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
18
16
14
12
11.65 [10.61–15.88]
11.73 [10.63–16.13]
13.30 [11.77–18.90]
11.64 [10.52–16.06]
11.60 [10.66–16.64]
11.61 [10.87–17.03]
64
Number of tasks

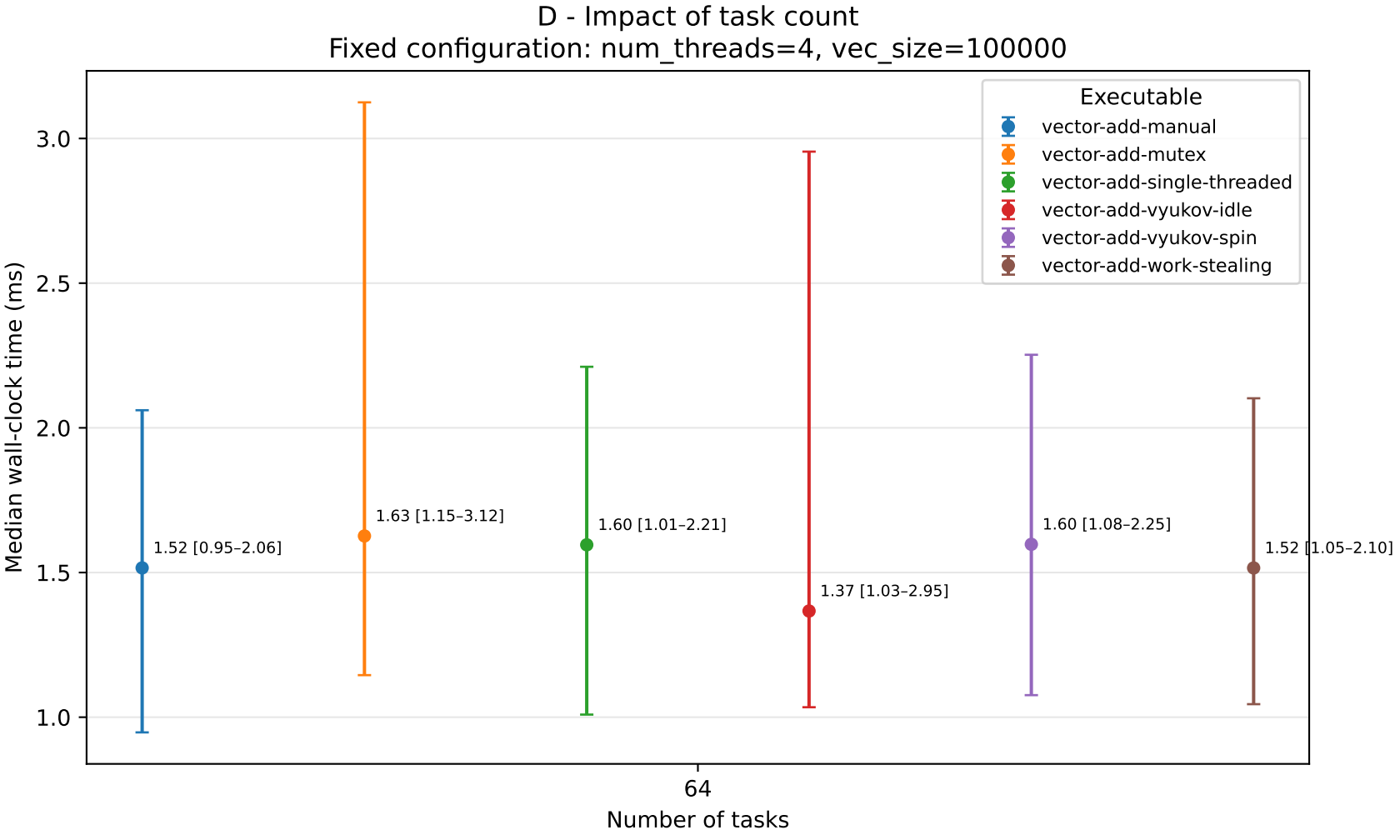

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=50000000
Median wall-clock time (ms)
28
26
24
22
20
18
64
Number of tasks
19.96 [18.22–26.48]
20.14 [18.93–27.92]
22.76 [21.13–27.88]
20.22 [19.03–26.60]
20.17 [18.74–26.63]
20.22 [18.73–25.55]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing

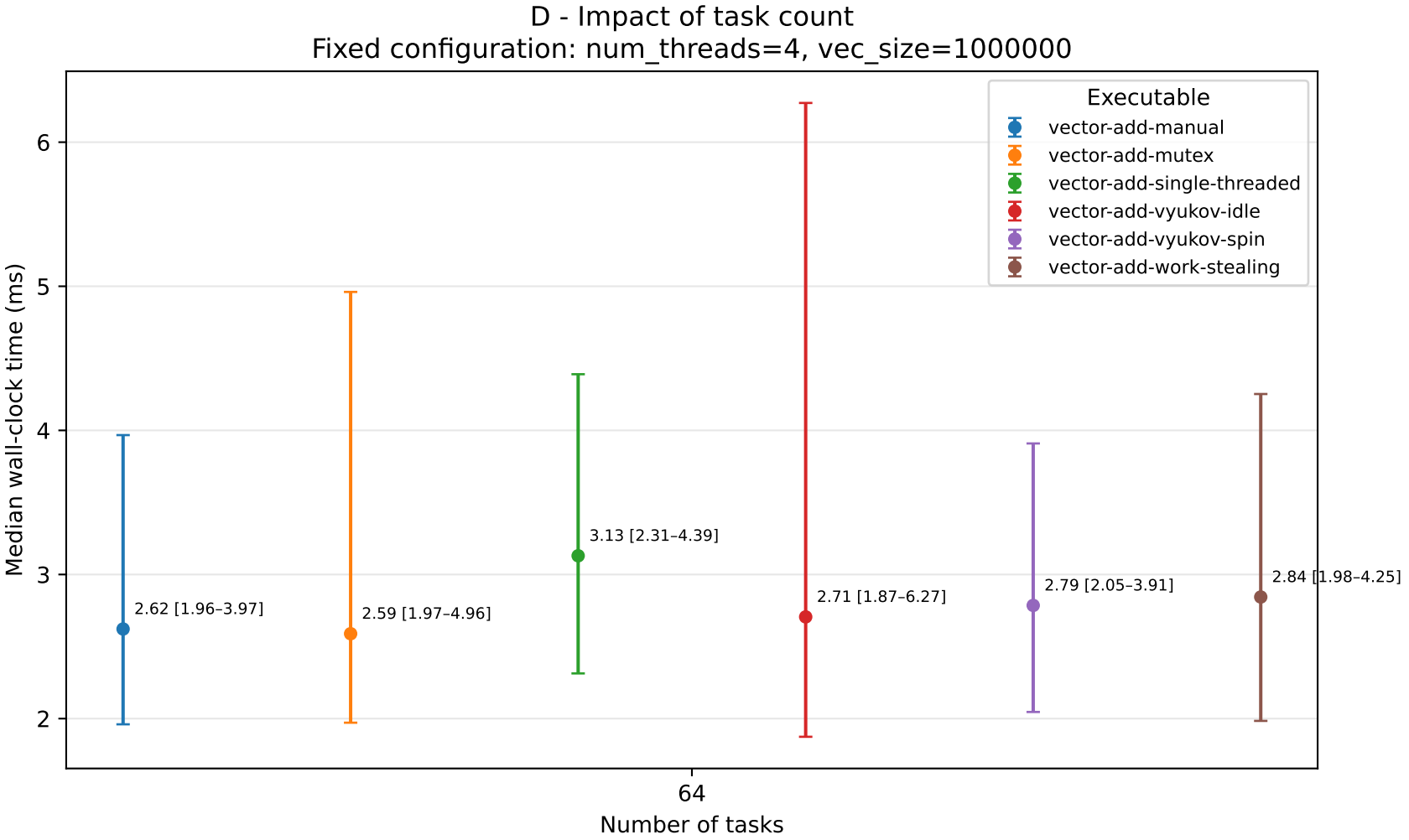

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=100000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
50
48
46
44
42
40
38
36
37.14 [34.97–45.83]
37.48 [35.42–44.12]
41.97 [39.55–51.11]
37.40 [35.32–44.16]
37.33 [35.24–43.50]
37.22 [35.60–42.79]
64
Number of tasks

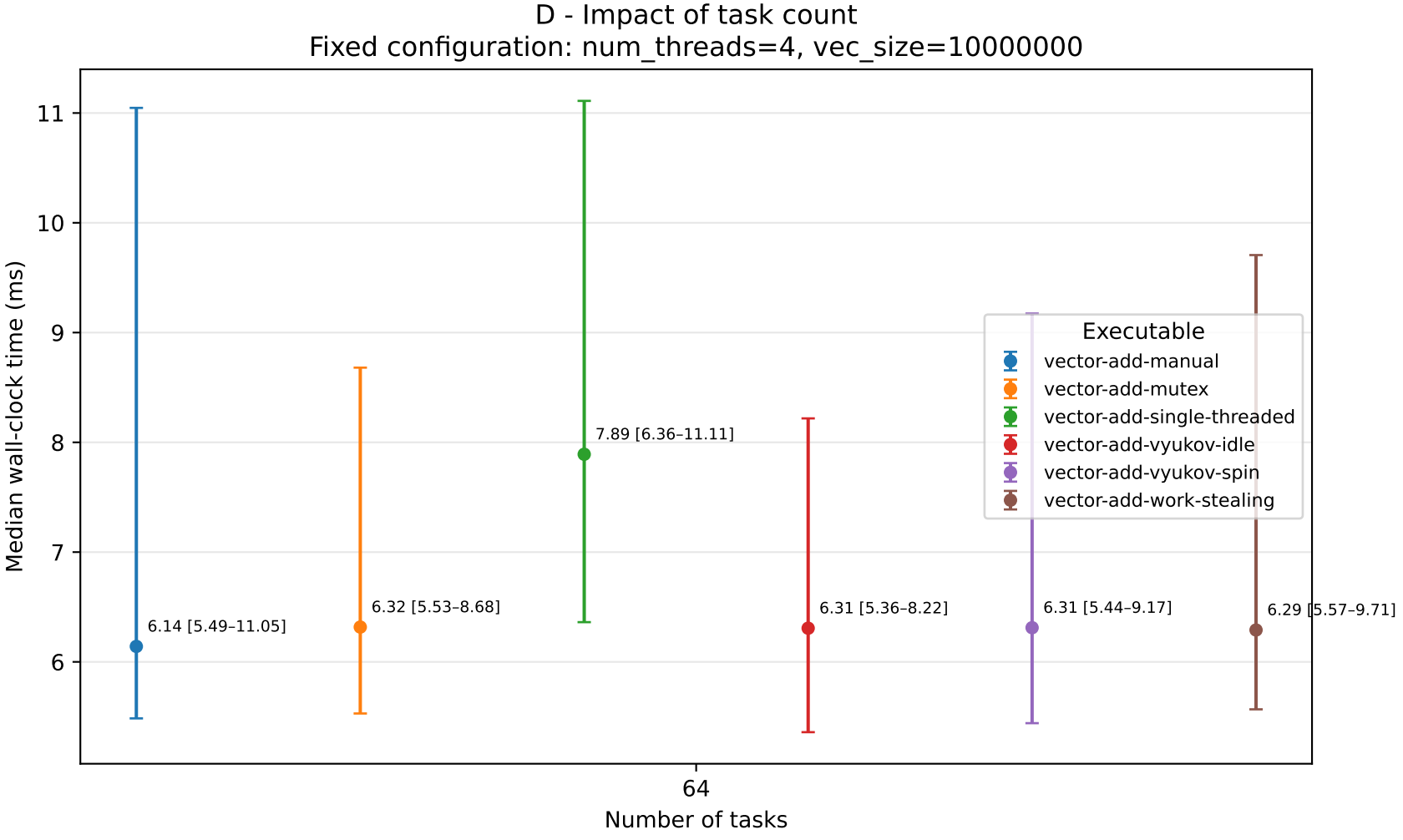

D - Impact of task count
Fixed configuration: num_threads=2, vec_size=200000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
95
90
85
80
75
70
65
72.28 [67.56–80.55]
71.85 [67.79–79.78]
81.96 [77.57–93.26]
71.27 [66.74–83.06]
71.73 [66.73–81.54]
71.86 [65.76–96.88]
64
Number of tasks

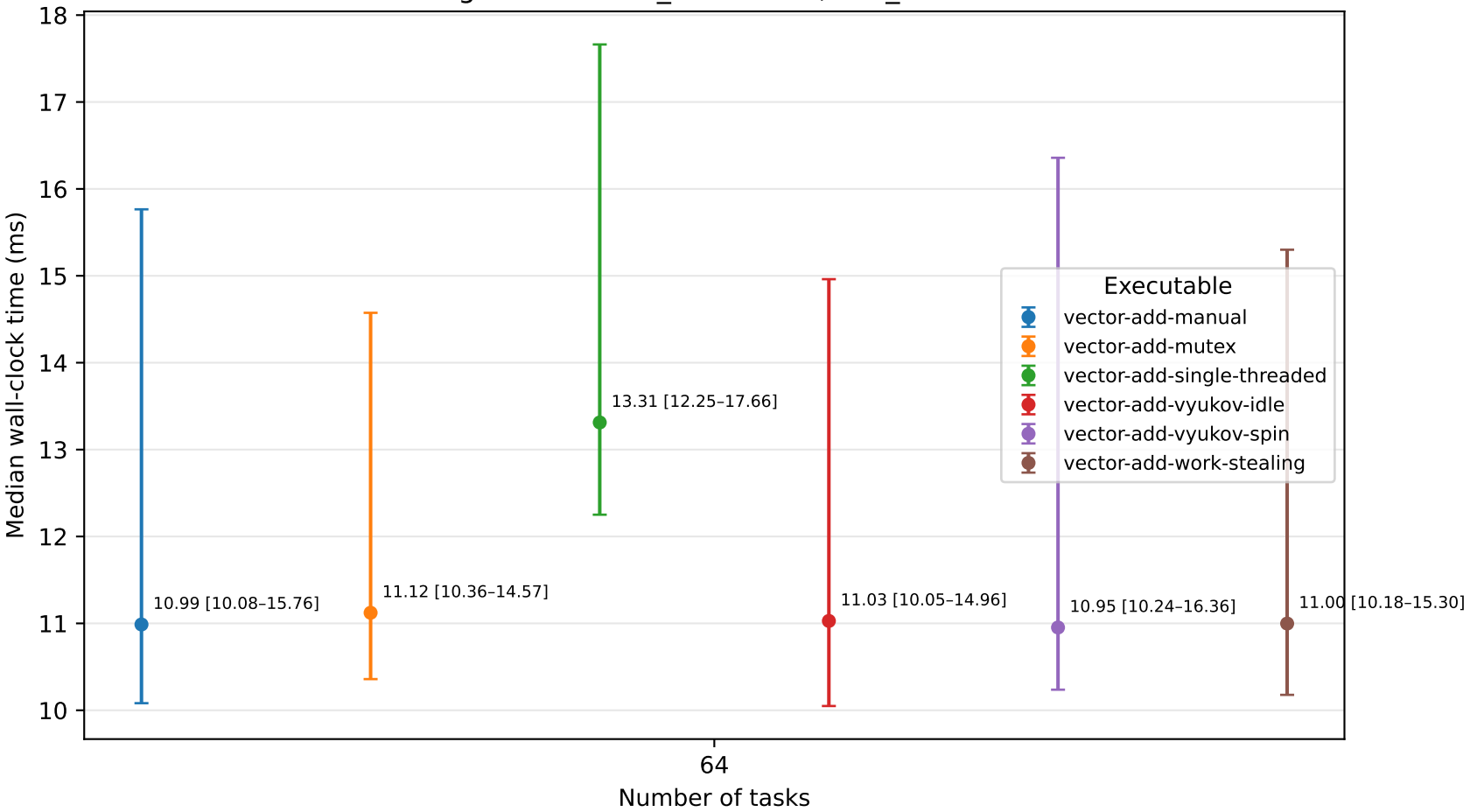

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=992
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
1.40 [0.85–3.04]
1.35 [0.89–2.43]
1.21 [0.76–2.41]
1.29 [0.88–2.61]
1.33 [0.88–2.63]
1.39 [0.91–1.93]
64
Number of tasks

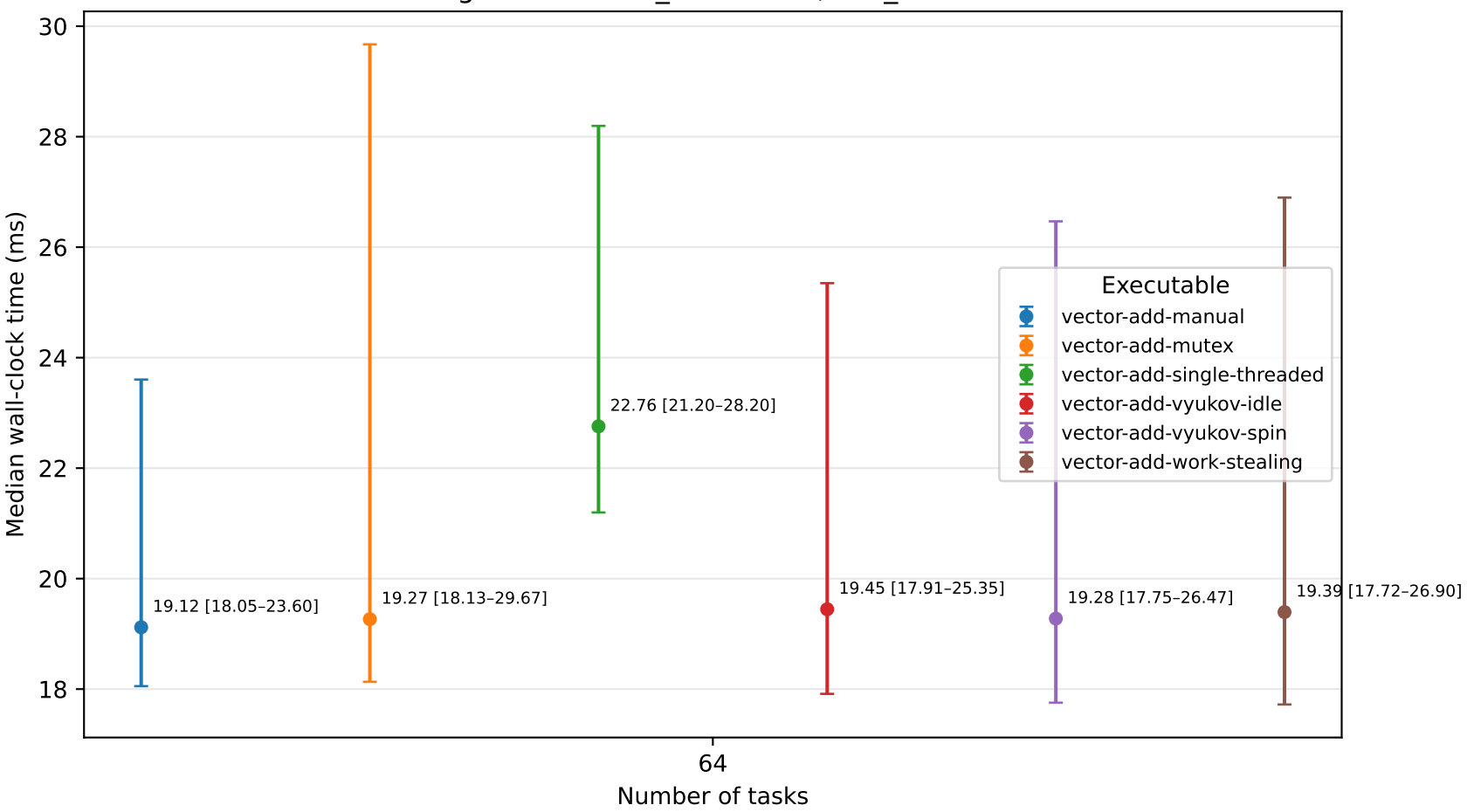

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=10016
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.39 [0.84–2.09]
1.32 [0.94–2.91]
0.95 [0.76–1.69]
1.49 [0.95–2.50]
1.40 [0.92–2.30]
1.50 [0.94–2.73]
64
Number of tasks

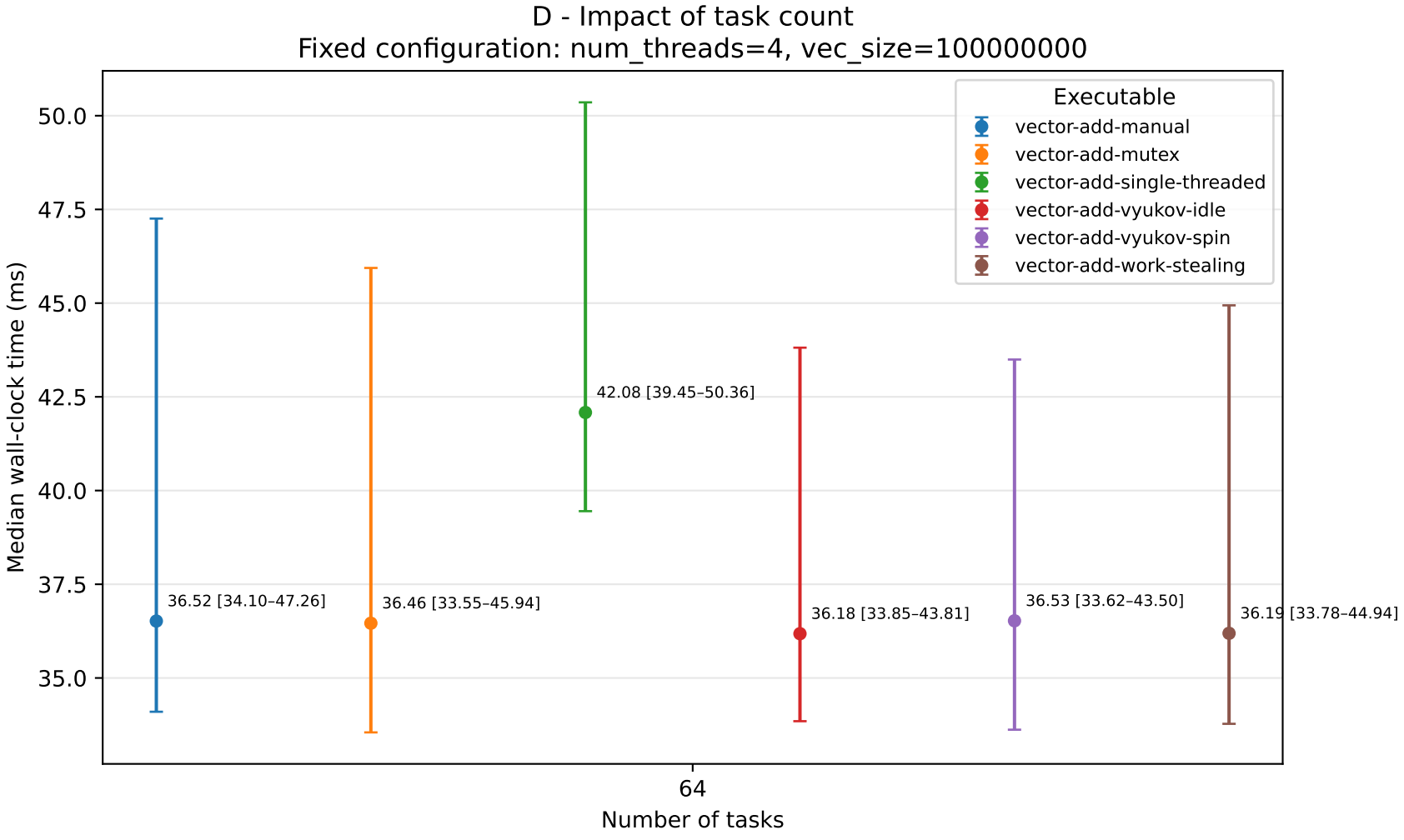

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=100000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.52 [0.95–2.06]
1.63 [1.15–3.12]
1.60 [1.01–2.21]
1.37 [1.03–2.95]
1.60 [1.08–2.25]
1.52 [1.05–2.10]
64
Number of tasks

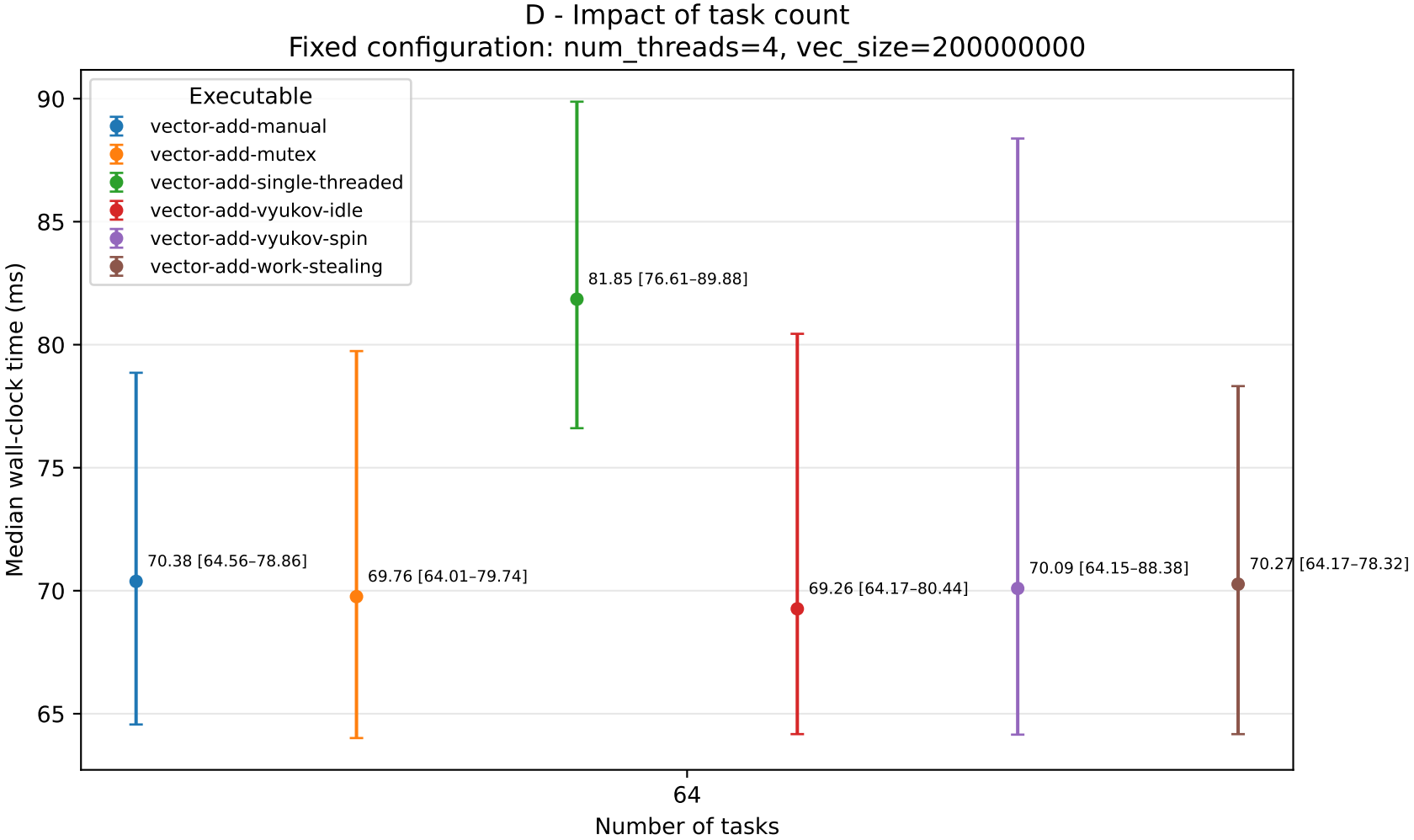

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=1000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
2
3
4
5
6
2.62 [1.96–3.97]
2.59 [1.97–4.96]
3.13 [2.31–4.39]
2.71 [1.87–6.27]
2.79 [2.05–3.91]
2.84 [1.98–4.25]
64
Number of tasks

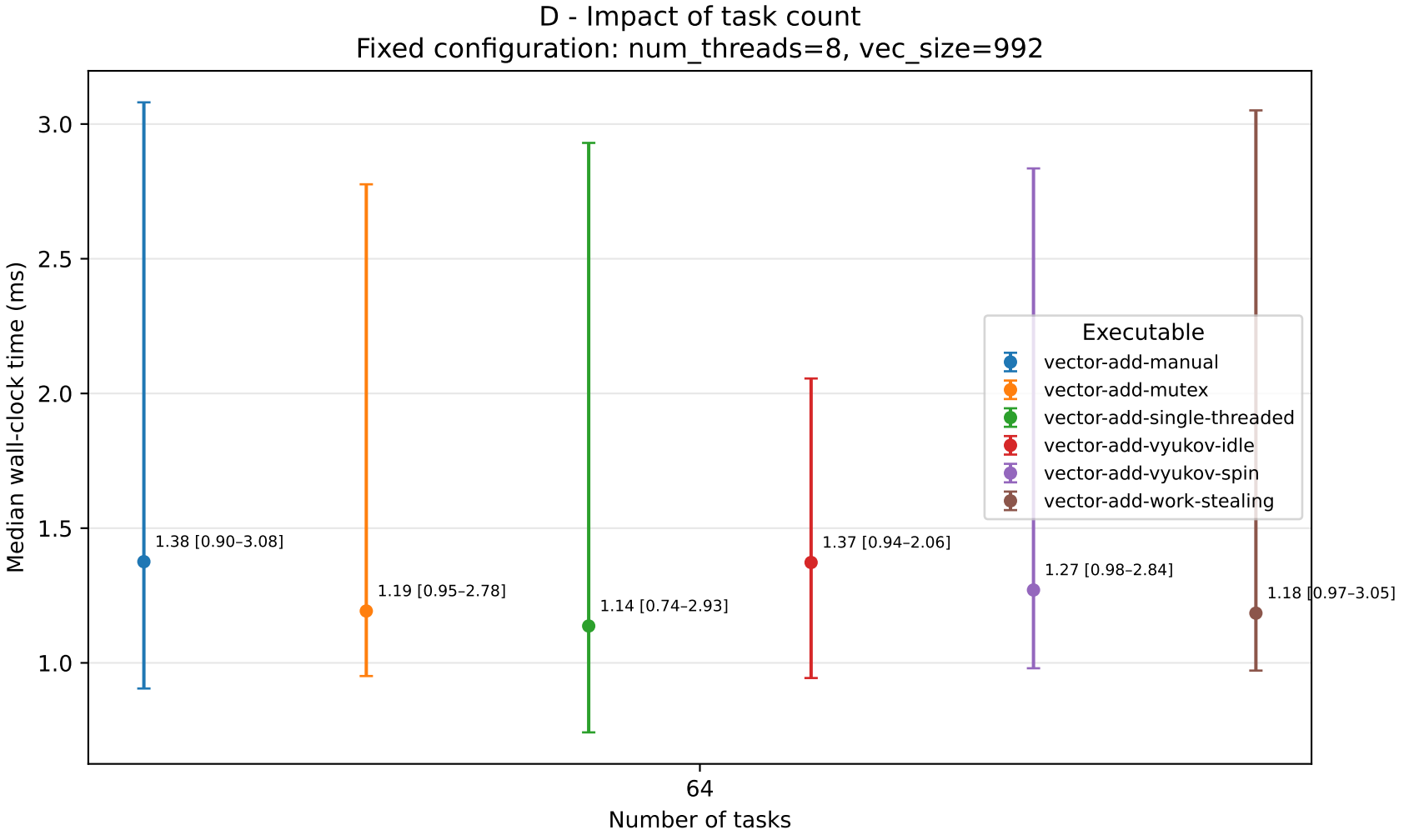

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=10000000
Median wall-clock time (ms)
11
10
9
8
7
6
6.14 [5.49–11.05]
6.32 [5.53–8.68]
7.89 [6.36–11.11]
6.31 [5.36–8.22]
6.31 [5.44–9.17]
6.29 [5.57–9.71]
64
Number of tasks
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing

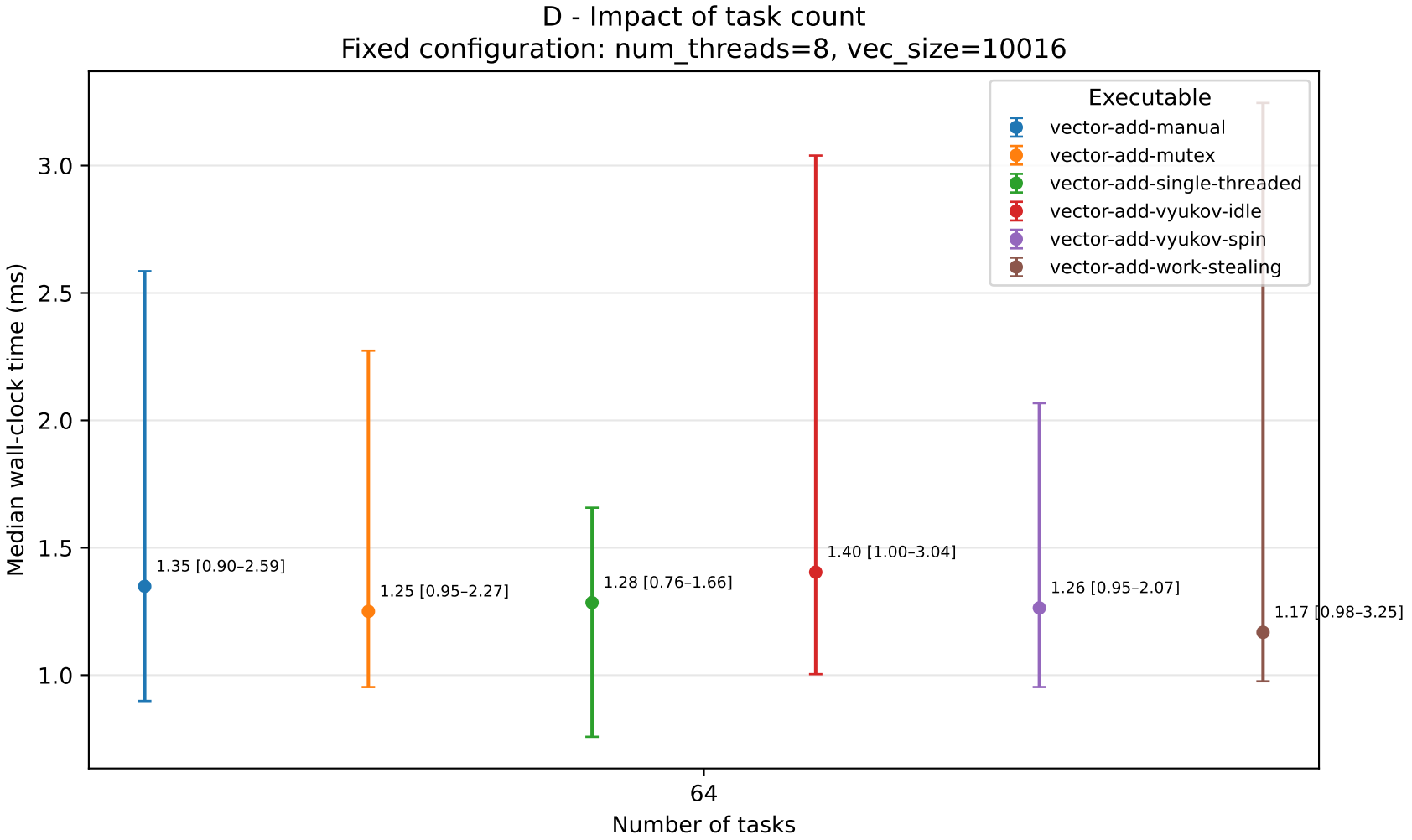

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=25000000
Median wall-clock time (ms)
18
17
16
15
14
13
12
11
10
64
Number of tasks
10.99 [10.08–15.76]
11.12 [10.36–14.57]
13.31 [12.25–17.66]
11.03 [10.05–14.96]
10.95 [10.24–16.36]
11.00 [10.18–15.30]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing

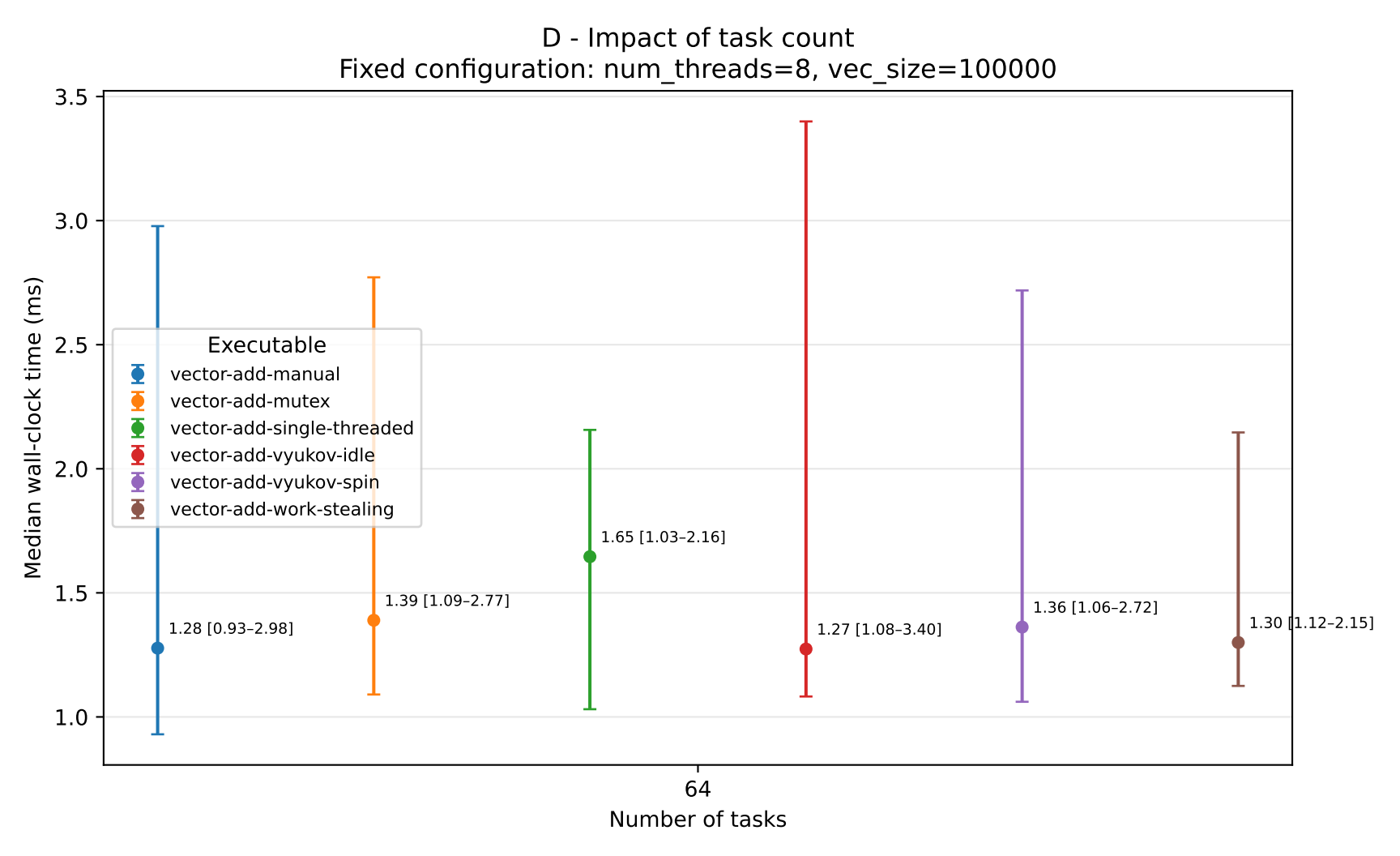

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=50000000
Median wall-clock time (ms)
30
28
26
24
22
20
18
19.12 [18.05–23.60]
19.27 [18.13–29.67]
22.76 [21.20–28.20]
19.45 [17.91–25.35]
19.28 [17.75–26.47]
19.39 [17.72–26.90]
64
Number of tasks
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing

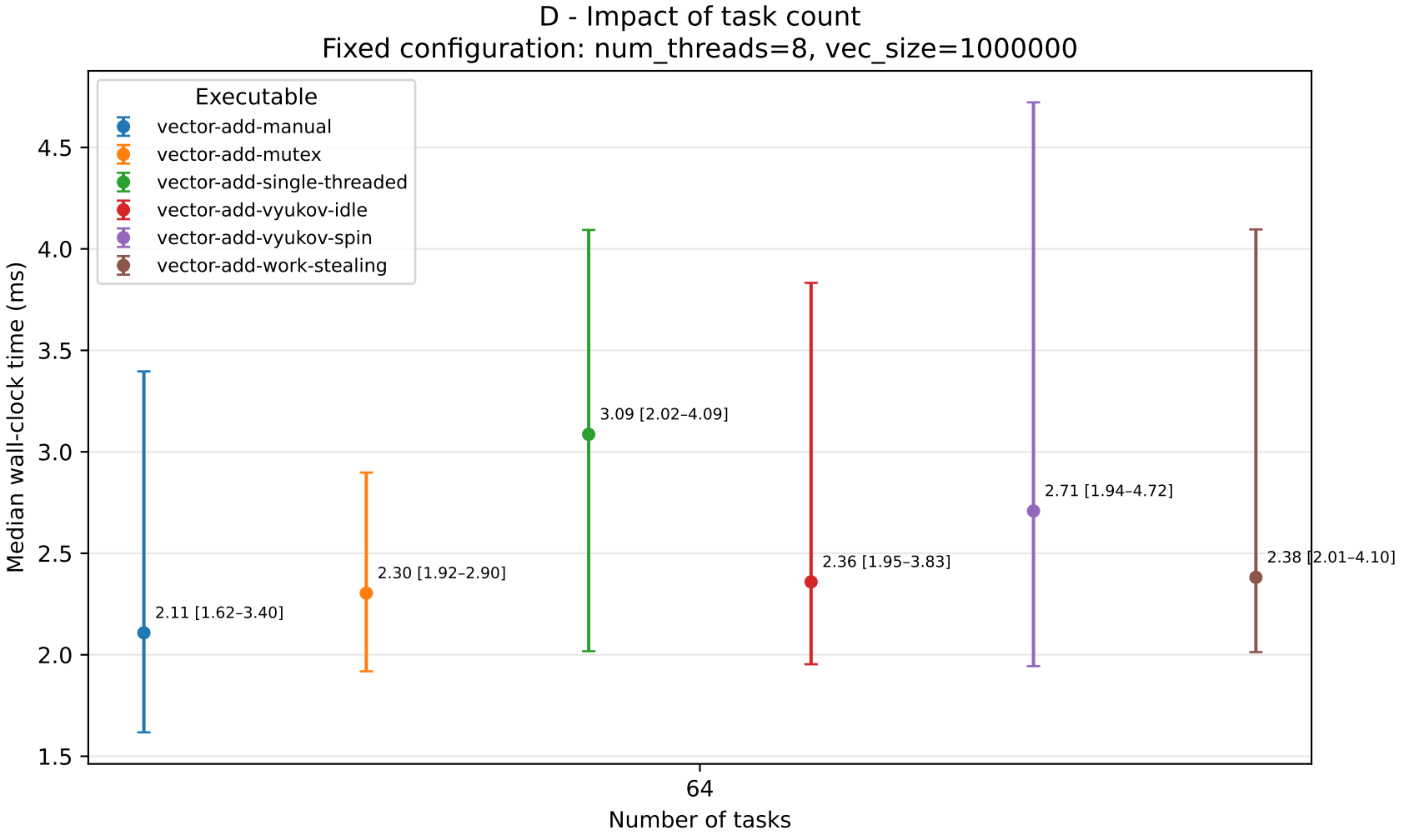

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=100000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
50.0
47.5
45.0
42.5
40.0
37.5
35.0
36.52 [34.10–47.26]
36.46 [33.55–45.94]
42.08 [39.45–50.36]
36.18 [33.85–43.81]
36.53 [33.62–43.50]
36.19 [33.78–44.94]
64
Number of tasks

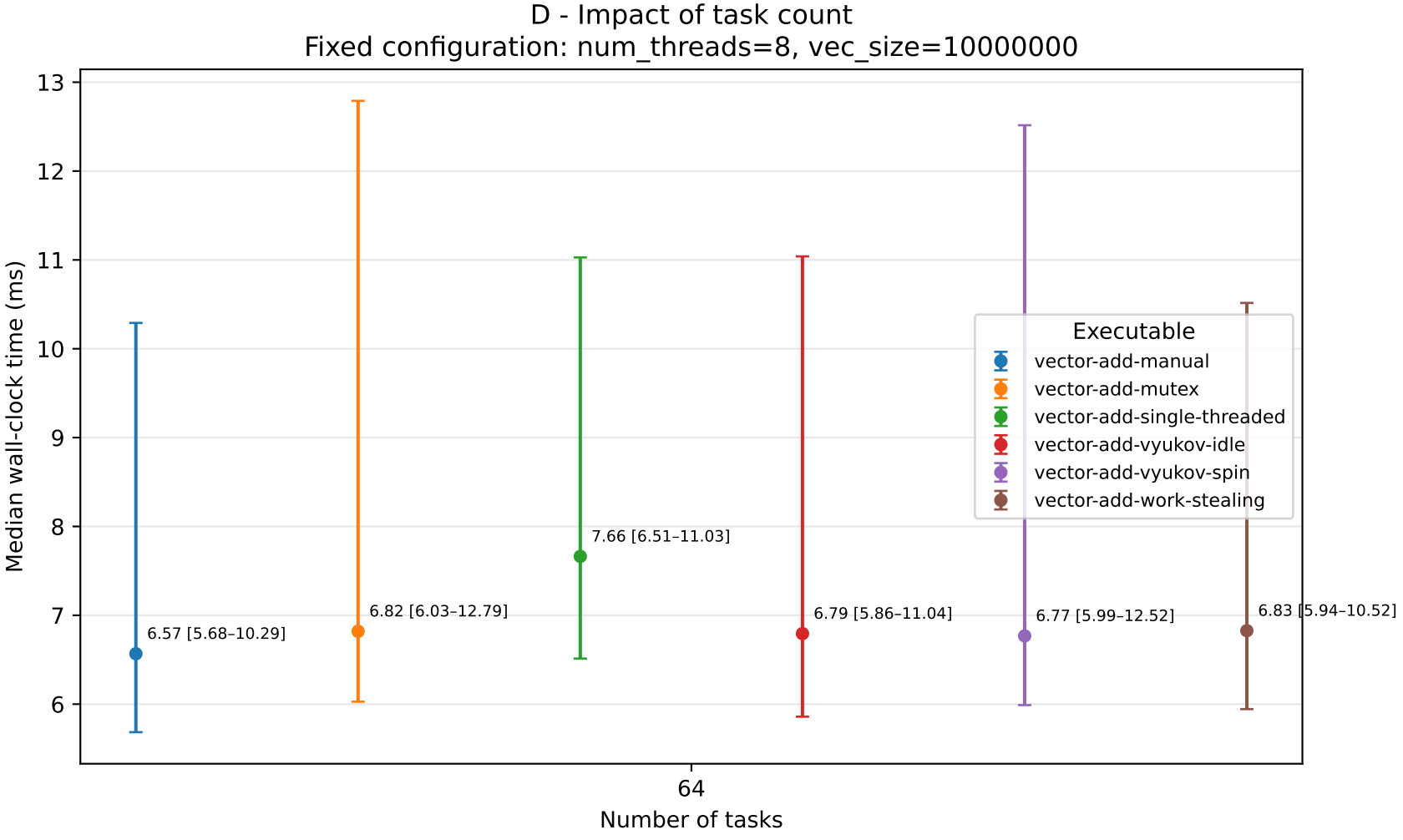

D - Impact of task count
Fixed configuration: num_threads=4, vec_size=200000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
90
85
80
75
70
65
70.38 [64.56–78.86]
69.76 [64.01–79.74]
81.85 [76.61–89.88]
69.26 [64.17–80.44]
70.09 [64.15–88.38]
70.27 [64.17–78.32]
64
Number of tasks

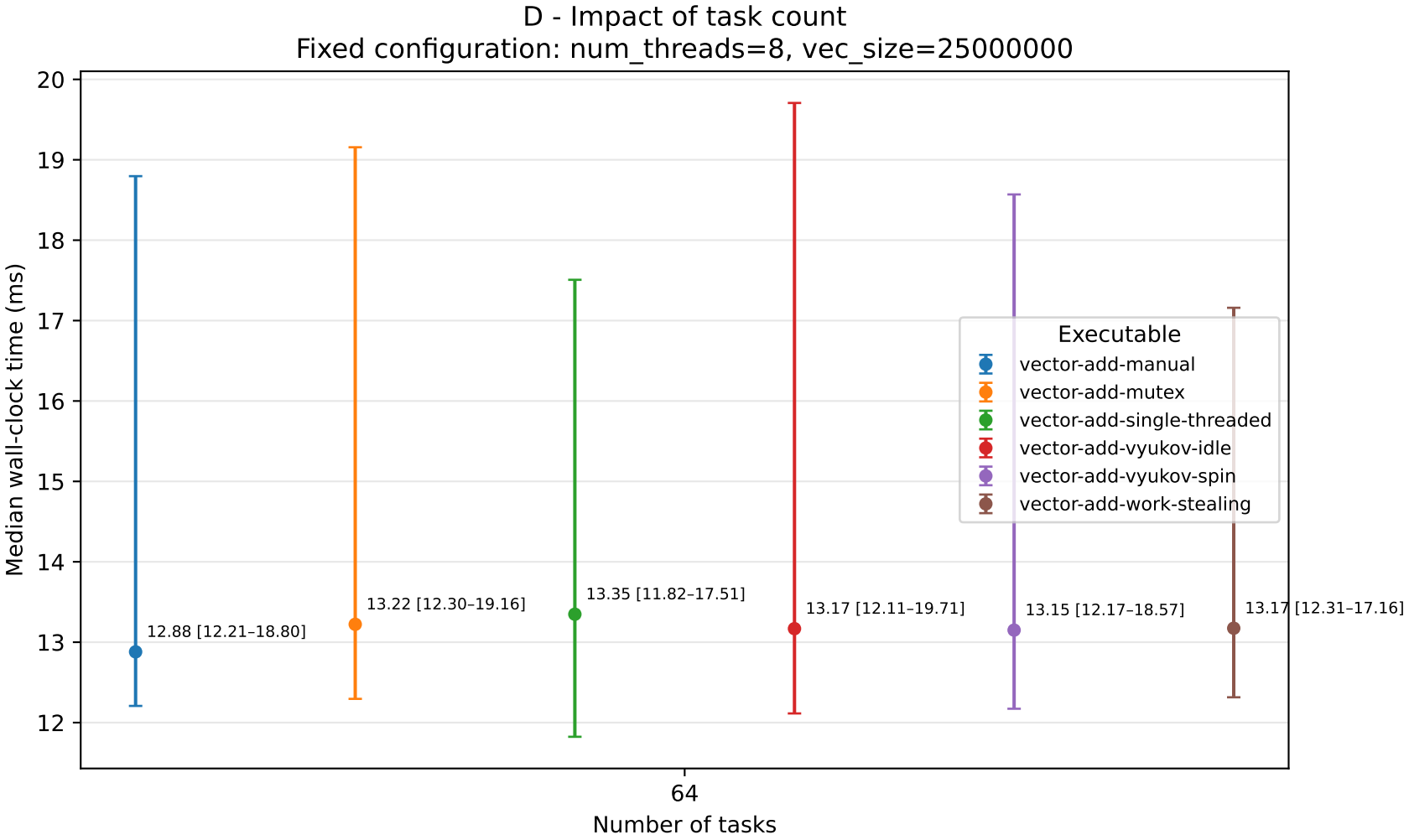

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=992
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.38 [0.90–3.08]
1.19 [0.95–2.78]
1.14 [0.74–2.93]
1.37 [0.94–2.06]
1.27 [0.98–2.84]
1.18 [0.97–3.05]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks

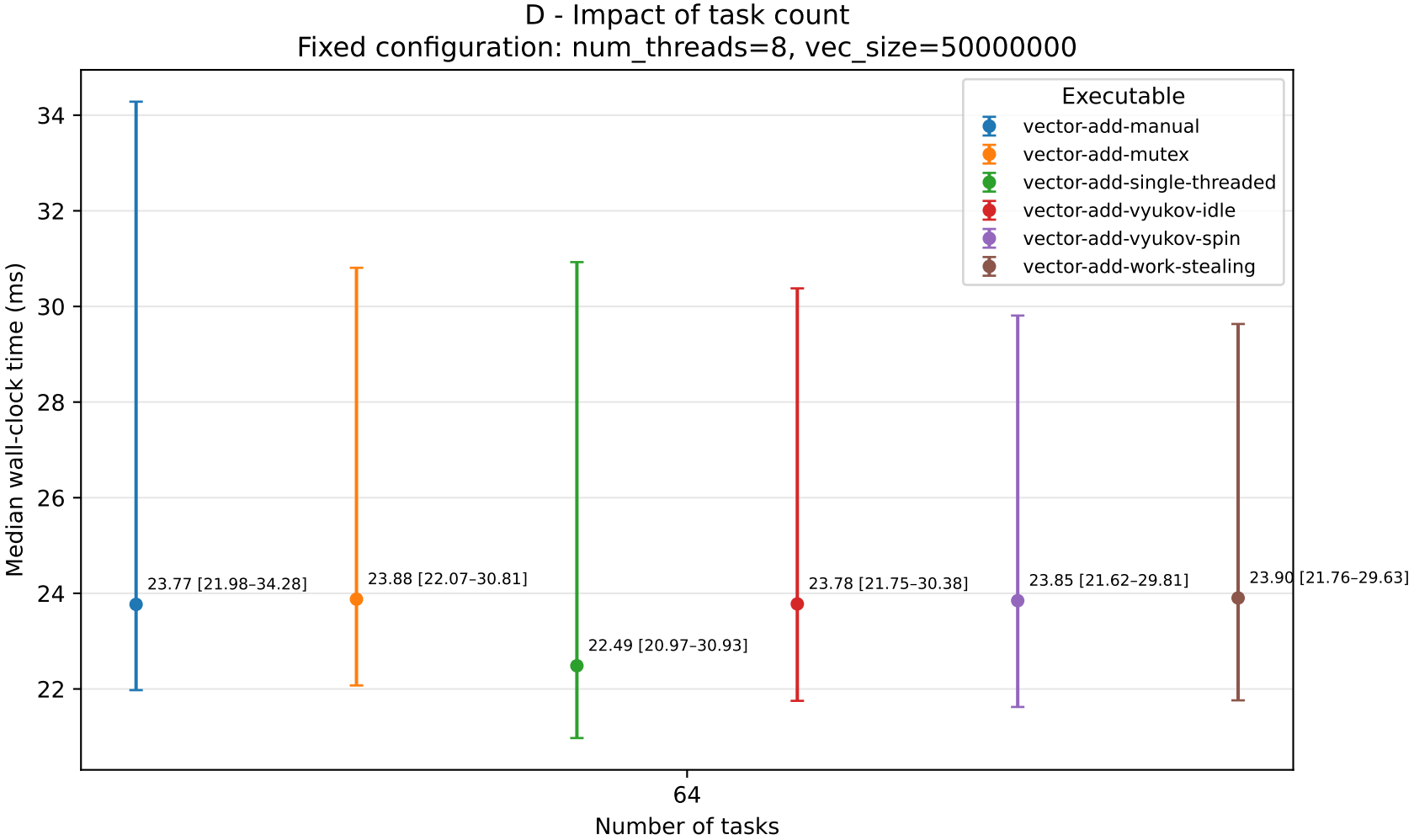

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=10016
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
3.0
2.5
2.0
1.5
1.0
1.35 [0.90–2.59]
1.25 [0.95–2.27]
1.28 [0.76–1.66]
1.40 [1.00–3.04]
1.26 [0.95–2.07]
1.17 [0.98–3.25]
64
Number of tasks

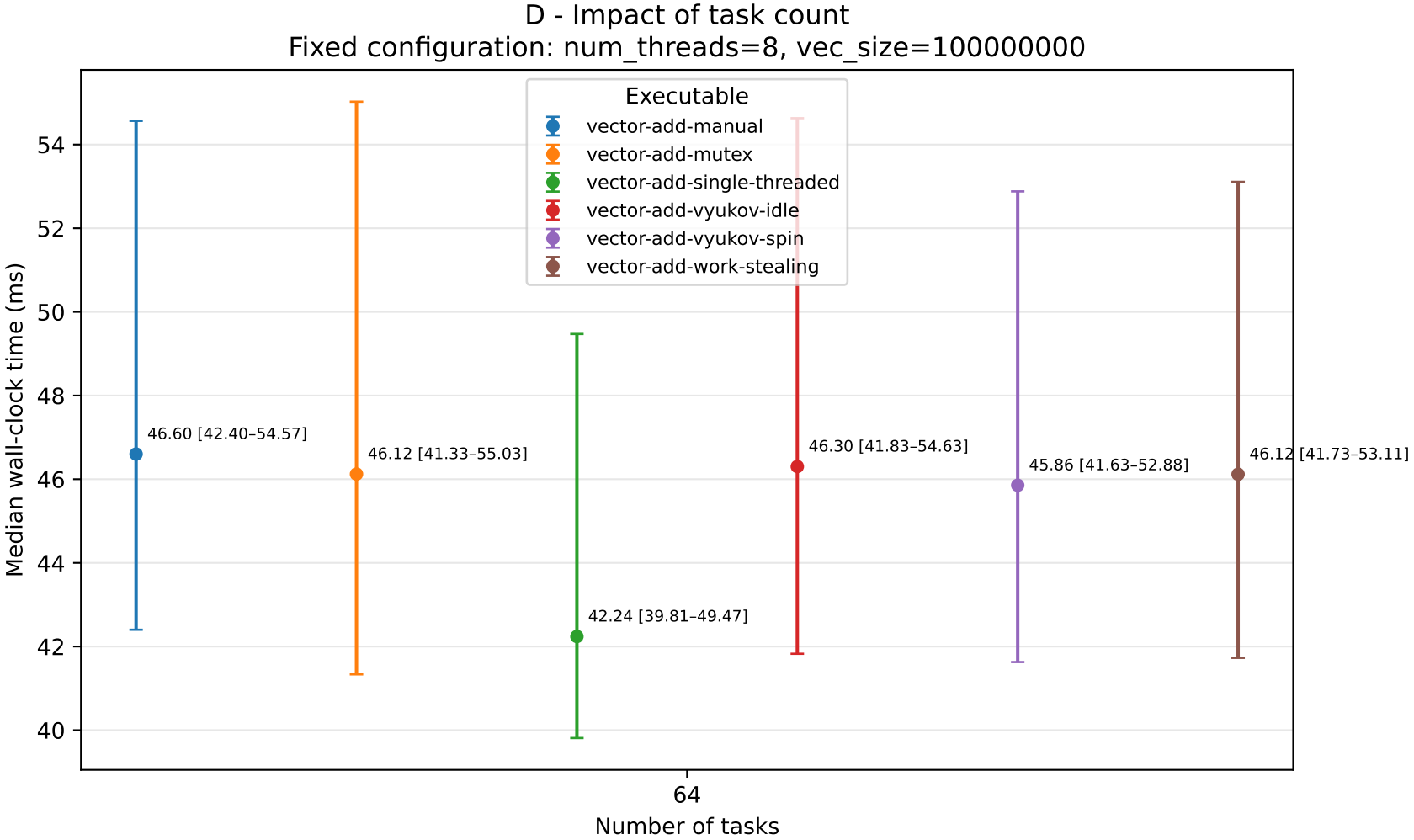

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=100000
Median wall-clock time (ms)
3.5
3.0
2.5
2.0
1.5
1.0
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
1.28 [0.93–2.98]
1.39 [1.09–2.77]
1.65 [1.03–2.16]
1.27 [1.08–3.40]
1.36 [1.06–2.72]
1.30 [1.12–2.15]
64
Number of tasks

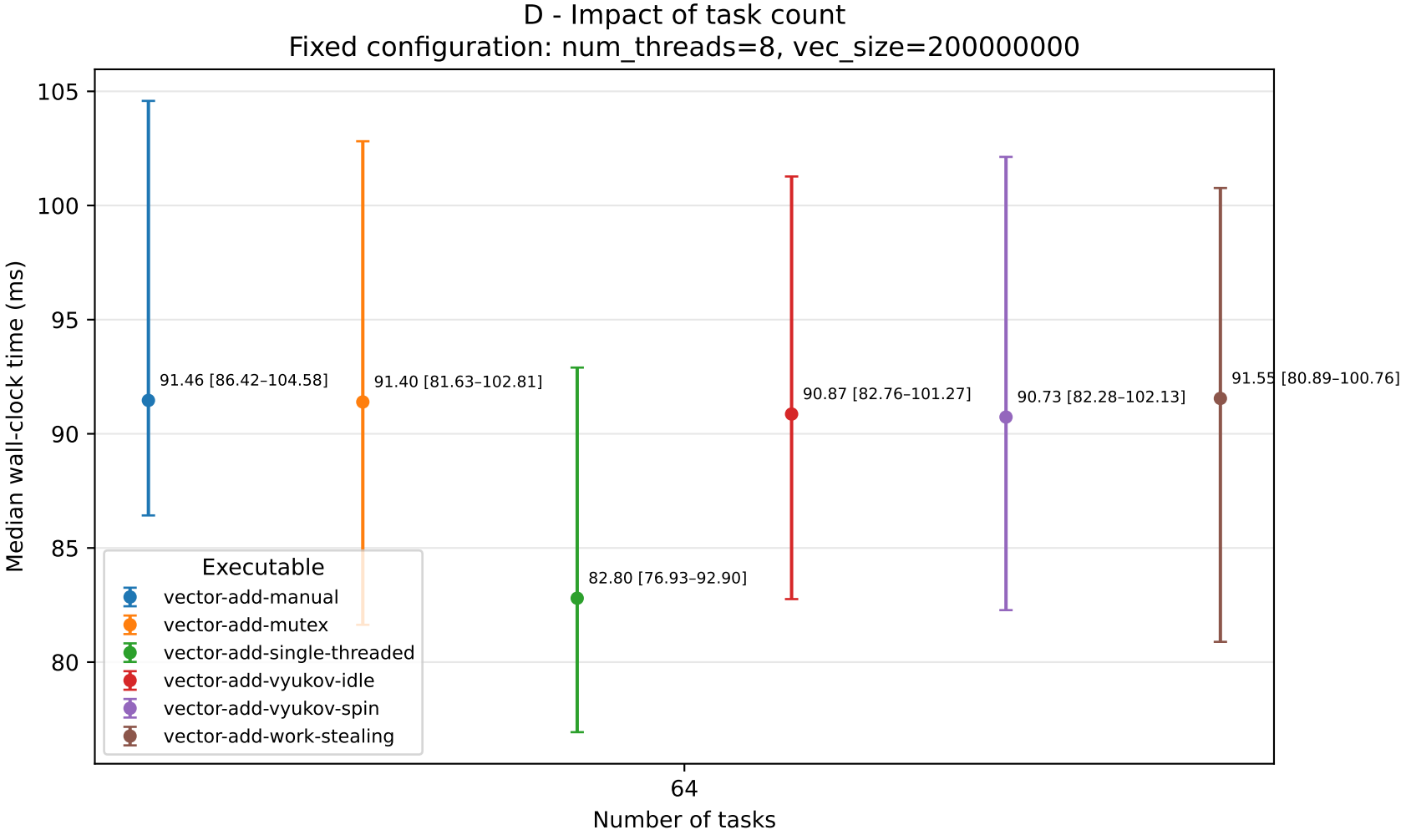

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=1000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
2.11 [1.62–3.40]
2.30 [1.92–2.90]
3.09 [2.02–4.09]
2.36 [1.95–3.83]
2.71 [1.94–4.72]
2.38 [2.01–4.10]
64
Number of tasks

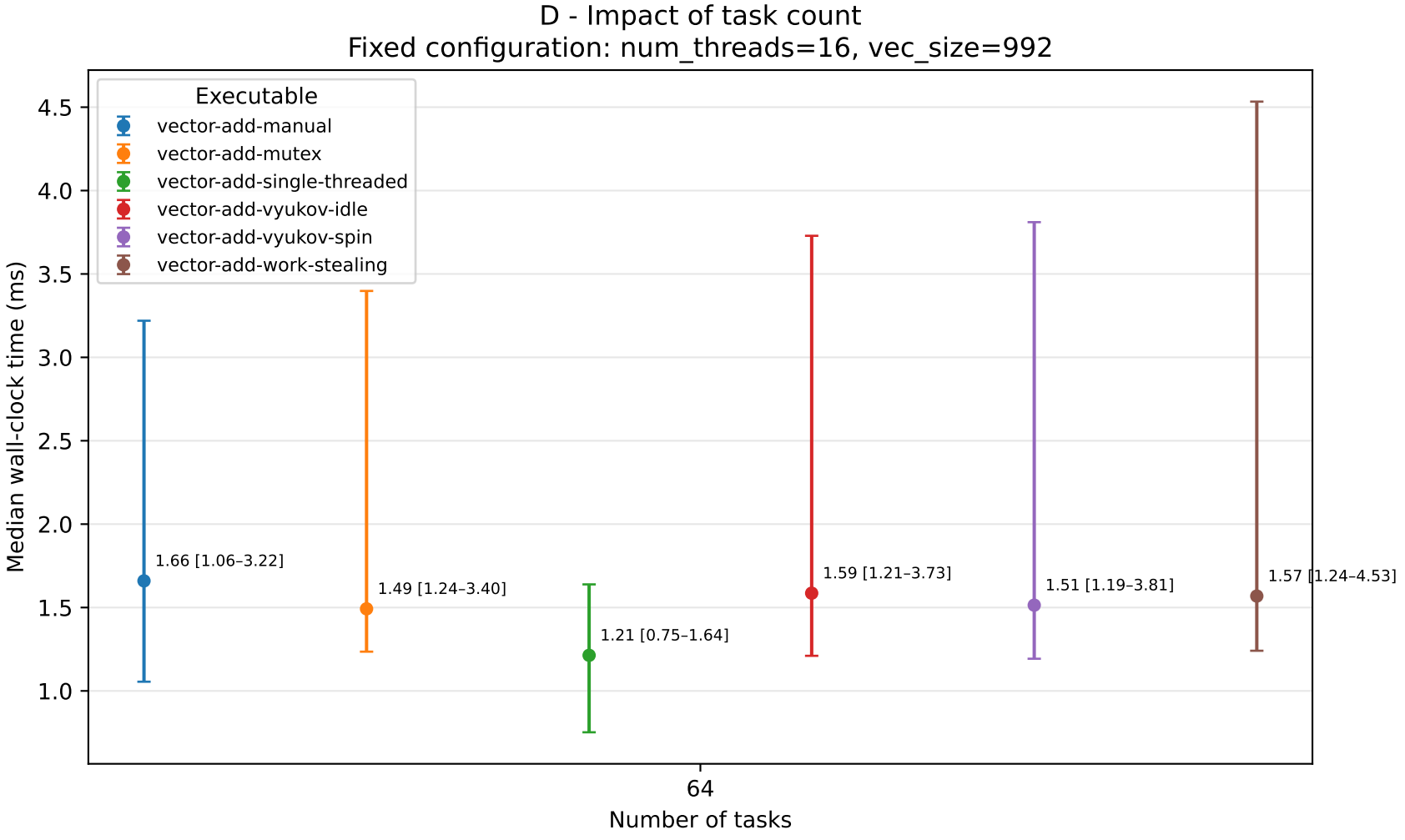

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=10000000
Median wall-clock time (ms)
13
12
11
10
9
8
7
6
64
Number of tasks
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
6.57 [5.68–10.29]
6.82 [6.03–12.79]
7.66 [6.51–11.03]
6.79 [5.86–11.04]
6.77 [5.99–12.52]
6.83 [5.94–10.52]

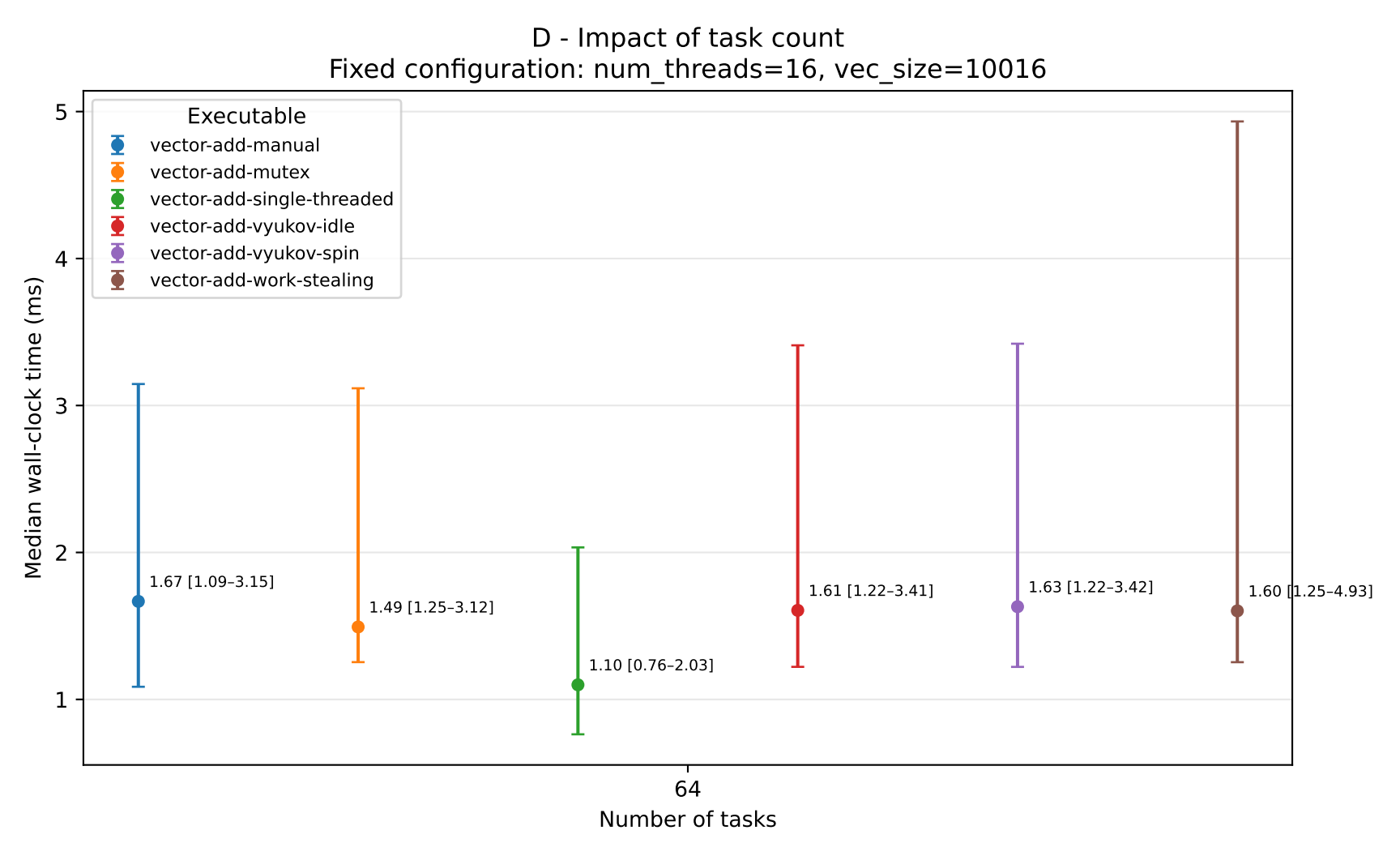

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=25000000
Median wall-clock time (ms)
20
19
18
17
16
15
14
13
12
12.88 [12.21–18.80]
13.22 [12.30–19.16]
13.35 [11.82–17.51]
13.17 [12.11–19.71]
13.15 [12.17–18.57]
13.17 [12.31–17.16]
64
Number of tasks
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing

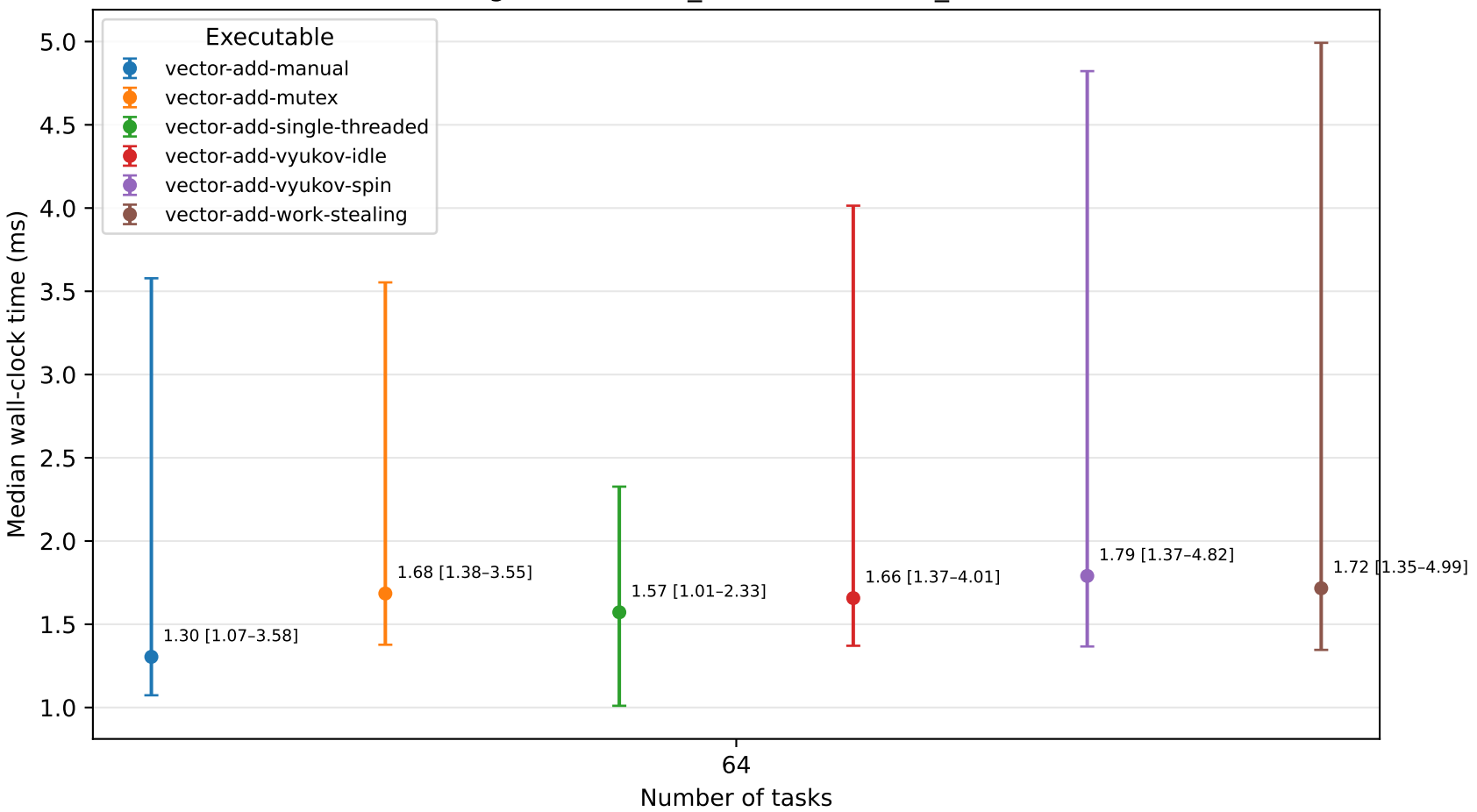

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=50000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
34
32
30
28
26
24
22
23.77 [21.98–34.28]
23.88 [22.07–30.81]
22.49 [20.97–30.93]
23.78 [21.75–30.38]
23.85 [21.62–29.81]
23.90 [21.76–29.63]
64
Number of tasks

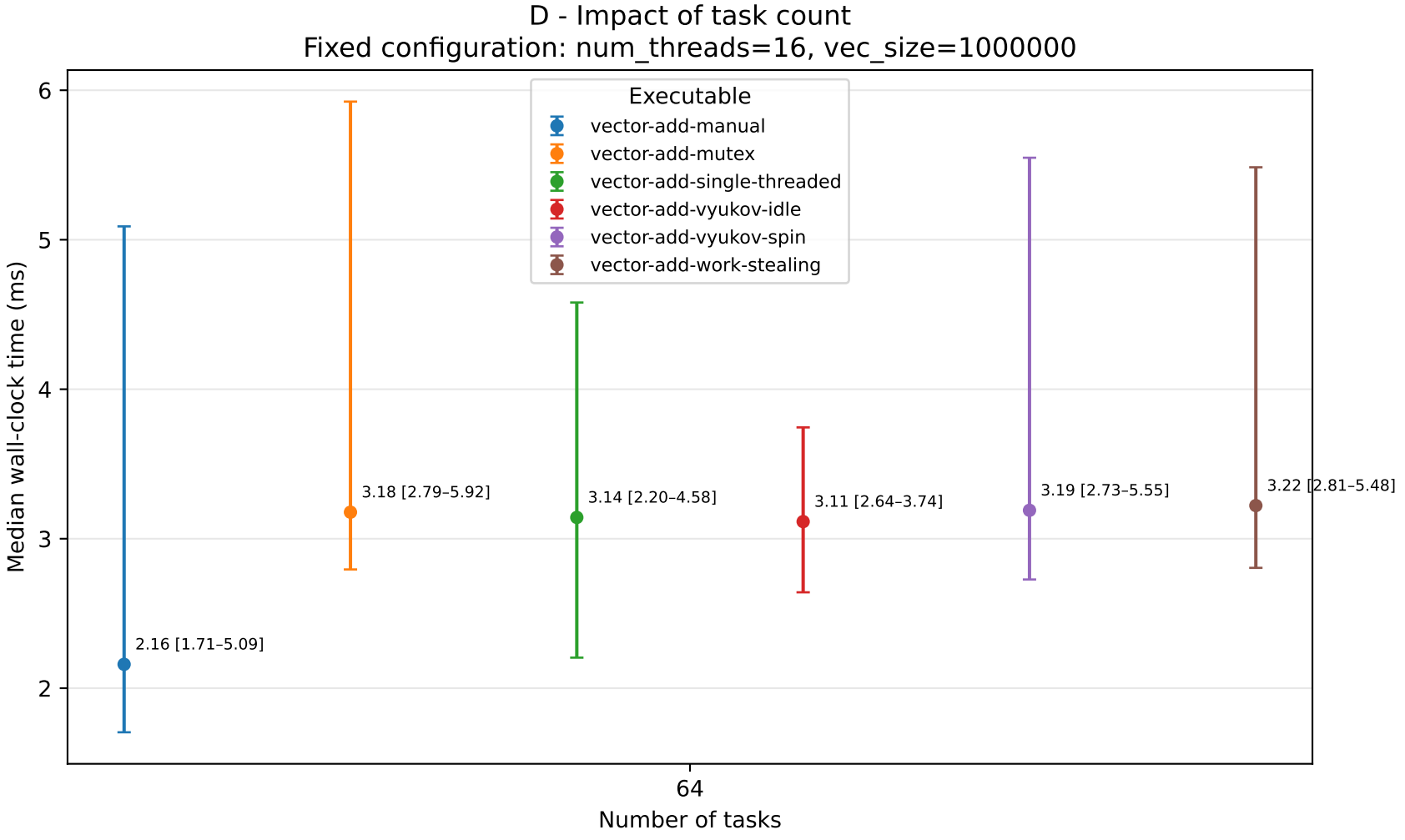

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=100000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
54
52
50
48
46
44
42
40
46.60 [42.40–54.57]
46.12 [41.33–55.03]
42.24 [39.81–49.47]
46.30 [41.83–54.63]
45.86 [41.63–52.88]
46.12 [41.73–53.11]
64
Number of tasks

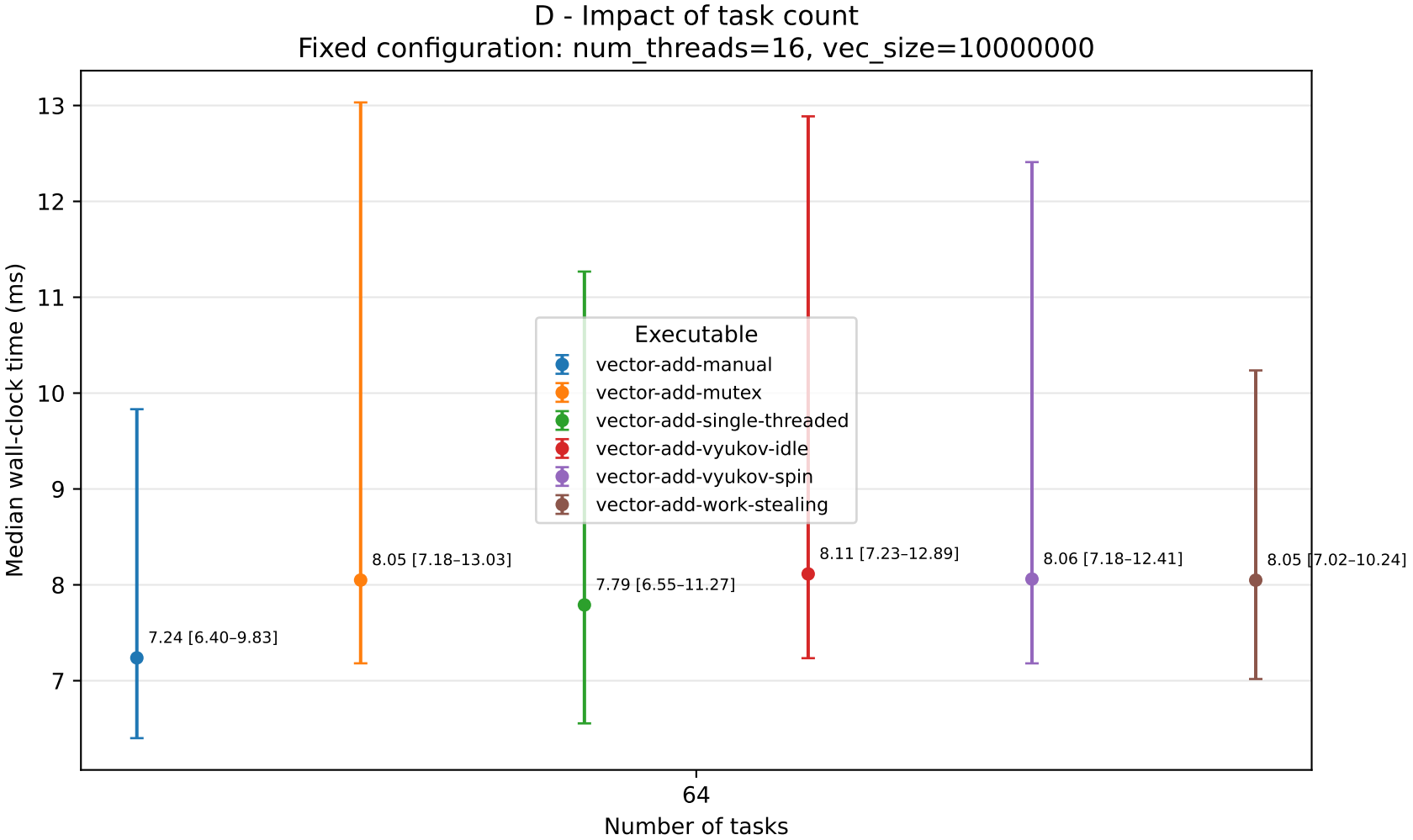

D - Impact of task count
Fixed configuration: num_threads=8, vec_size=200000000
Median wall-clock time (ms)
105
100
95
90
85
80
91.46 [86.42–104.58]
91.40 [81.63–102.81]
82.80 [76.93–92.90]
90.87 [82.76–101.27]
90.73 [82.28–102.13]
91.55 [80.89–100.76]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks

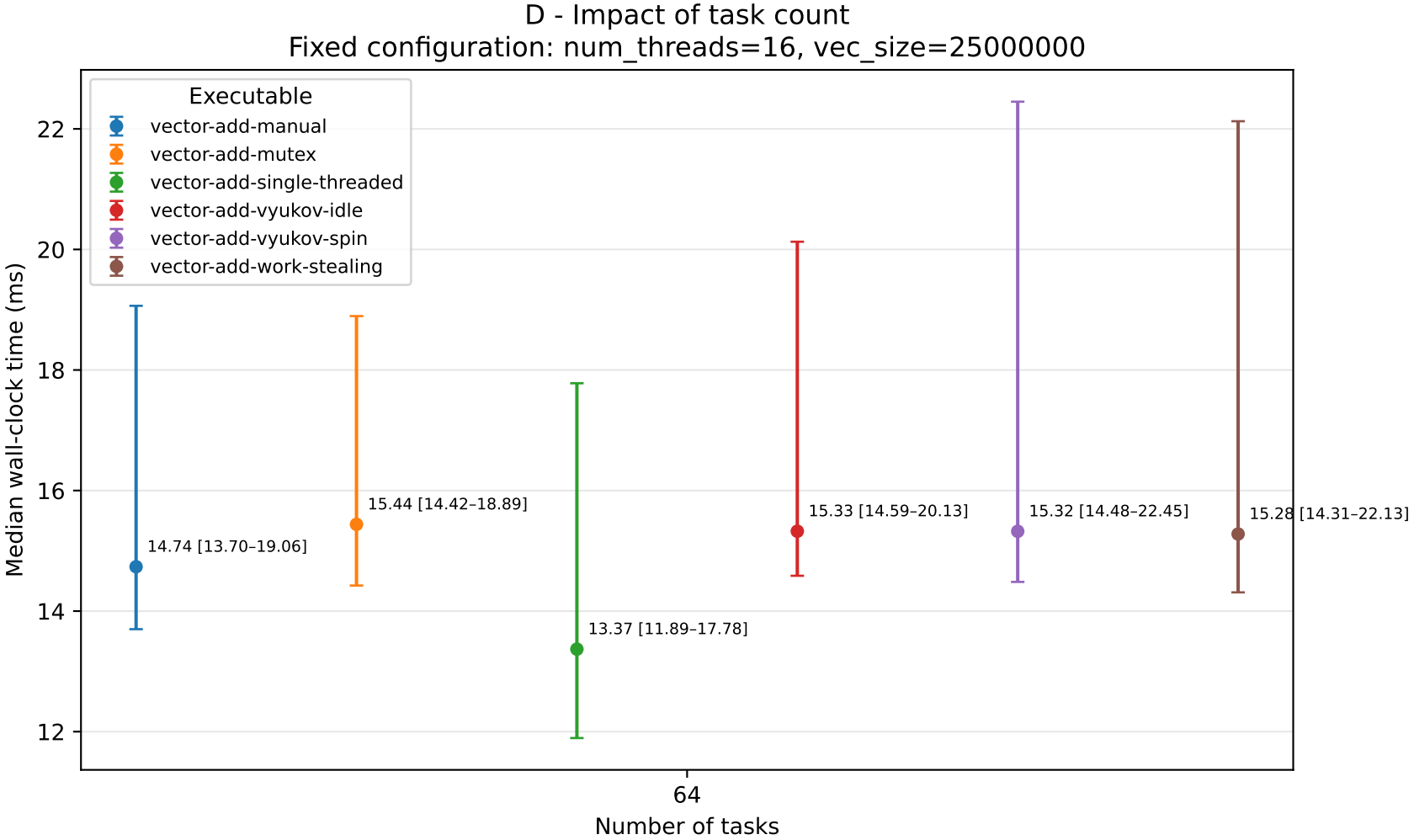

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=992
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.66 [1.06–3.22]
1.49 [1.24–3.40]
1.21 [0.75–1.64]
1.59 [1.21–3.73]
1.51 [1.19–3.81]
1.57 [1.24–4.53]
64
Number of tasks

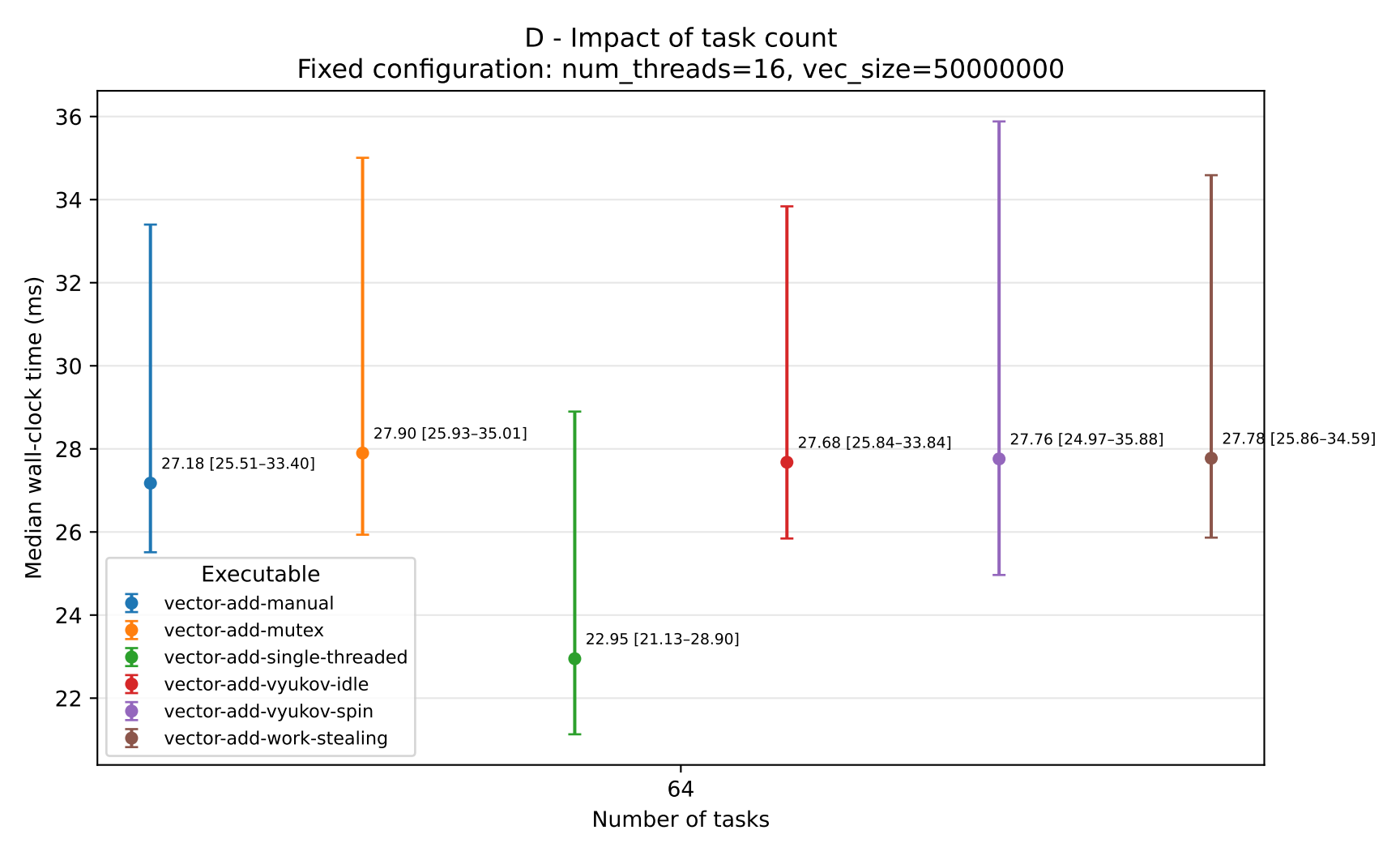

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=10016
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
5
4
3
2
1
1.67 [1.09–3.15]
1.49 [1.25–3.12]
1.10 [0.76–2.03]
1.61 [1.22–3.41]
1.63 [1.22–3.42]
1.60 [1.25–4.93]
64
Number of tasks

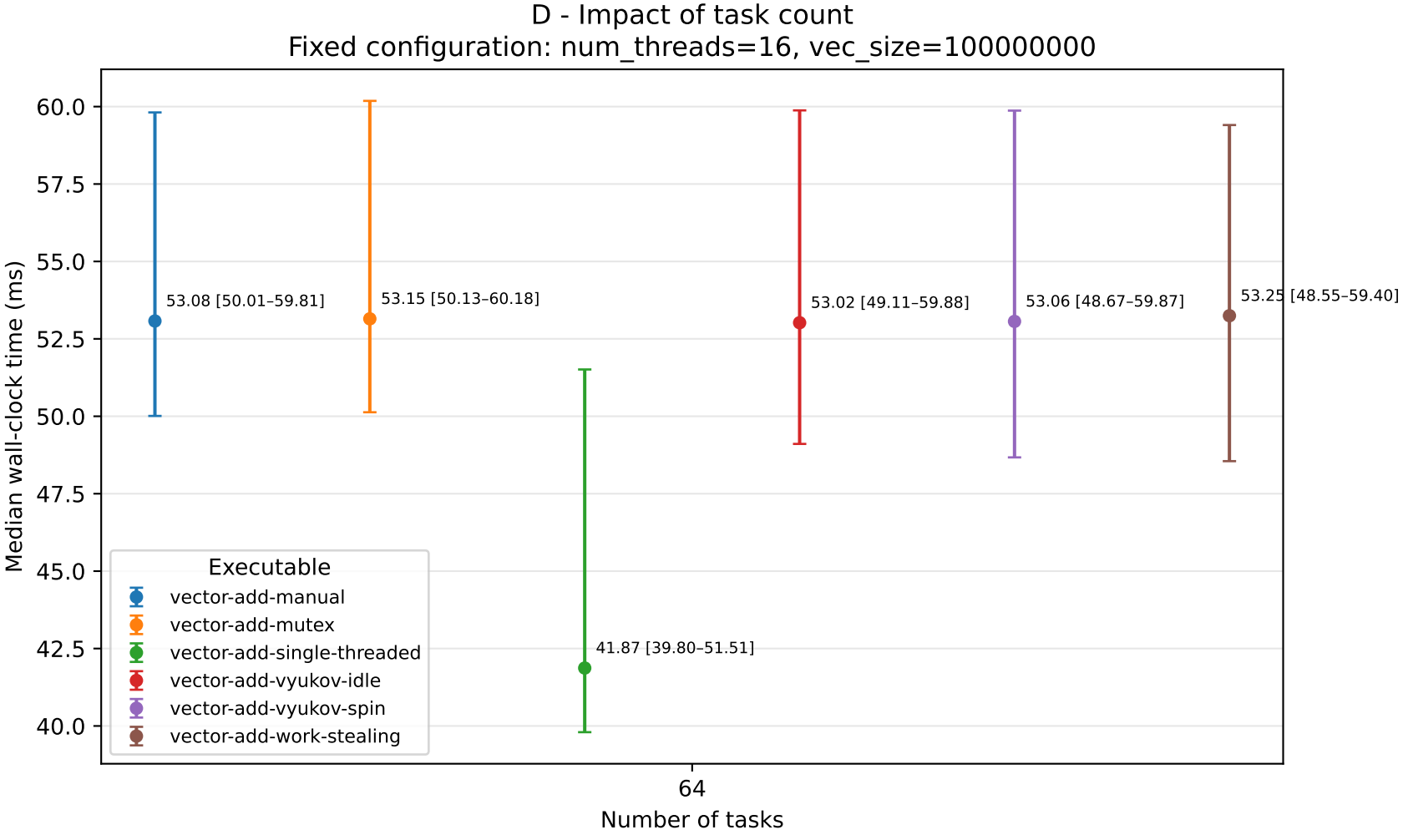

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=100000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.30 [1.07–3.58]
1.68 [1.38–3.55]
1.57 [1.01–2.33]
1.66 [1.37–4.01]
1.79 [1.37–4.82]
1.72 [1.35–4.99]
64
Number of tasks

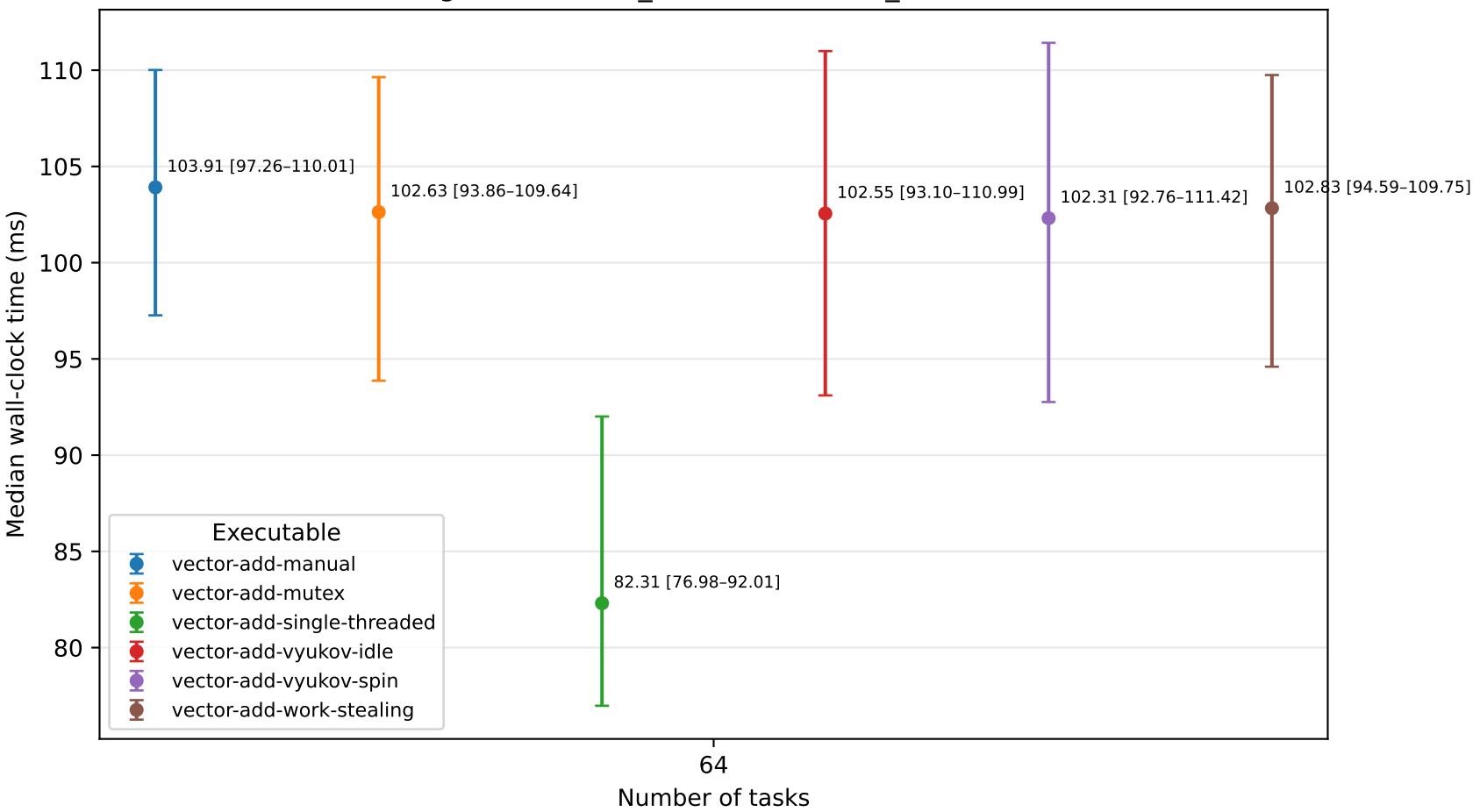

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=1000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
6
5
4
3
2
2.16 [1.71–5.09]
3.18 [2.79–5.92]
3.14 [2.20–4.58]
3.11 [2.64–3.74]
3.19 [2.73–5.55]
3.22 [2.81–5.48]
64
Number of tasks

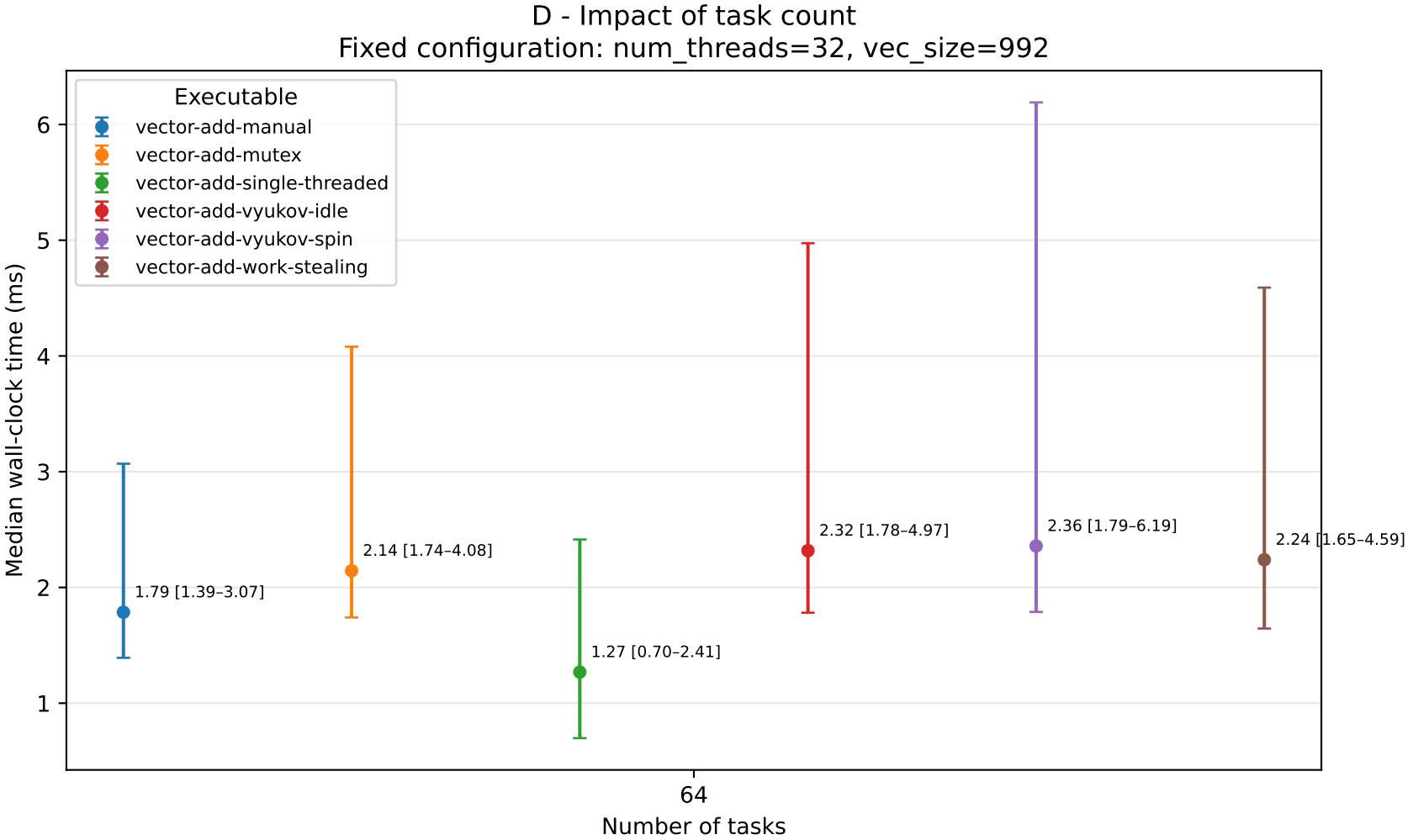

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=10000000
Median wall-clock time (ms)
13
12
11
10
9
8
7
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
7.24 [6.40–9.83]
8.05 [7.18–13.03]
7.79 [6.55–11.27]
8.11 [7.23–12.89]
8.06 [7.18–12.41]
8.05 [7.02–10.24]
64
Number of tasks

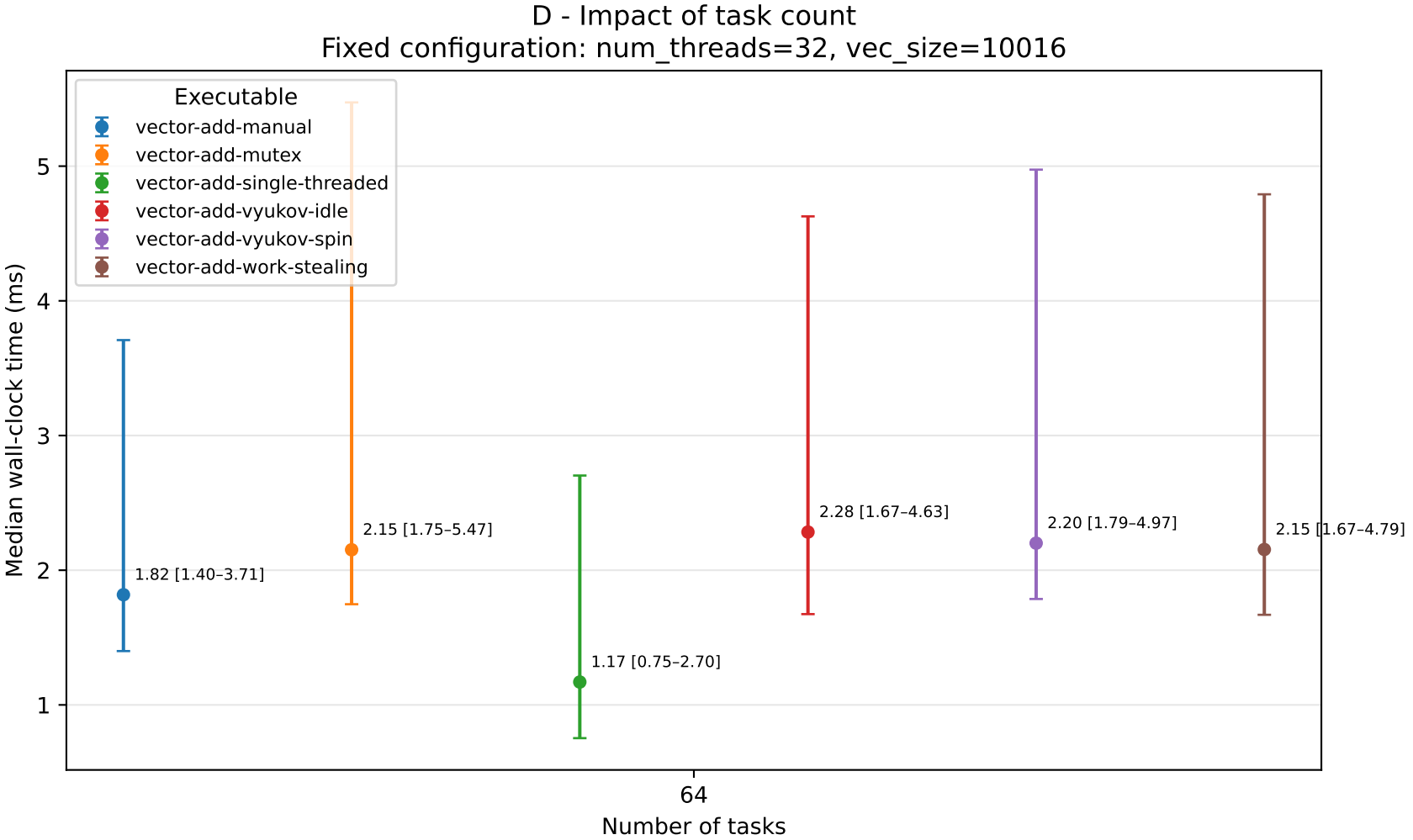

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=25000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
22
20
18
16
14
12
14.74 [13.70–19.06]
15.44 [14.42–18.89]
13.37 [11.89–17.78]
15.33 [14.59–20.13]
15.32 [14.48–22.45]
15.28 [14.31–22.13]
64
Number of tasks

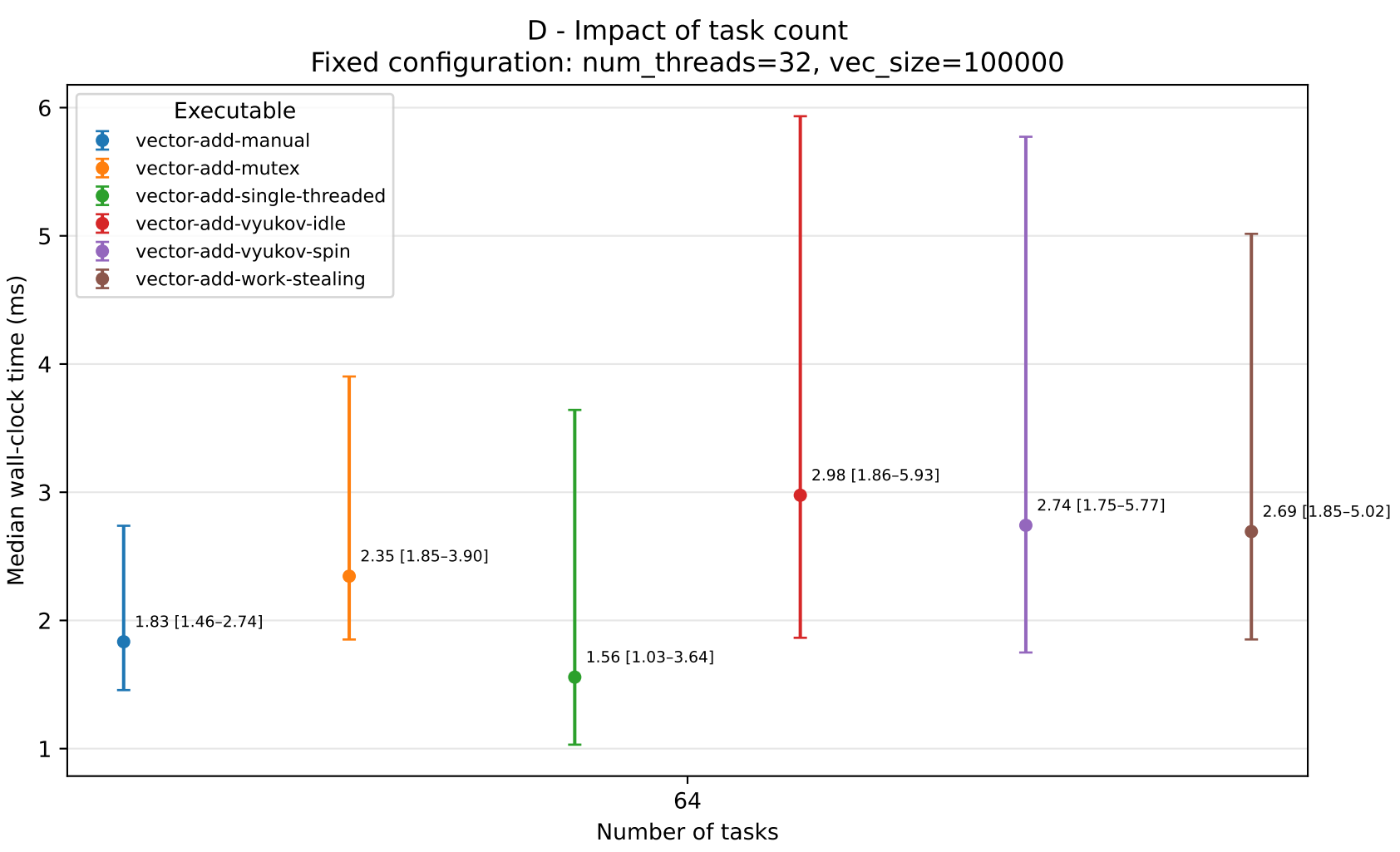

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=50000000
Median wall-clock time (ms)
36
34
32
30
28
26
24
22
27.18 [25.51–33.40]
27.90 [25.93–35.01]
22.95 [21.13–28.90]
27.68 [25.84–33.84]
27.76 [24.97–35.88]
27.78 [25.86–34.59]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks

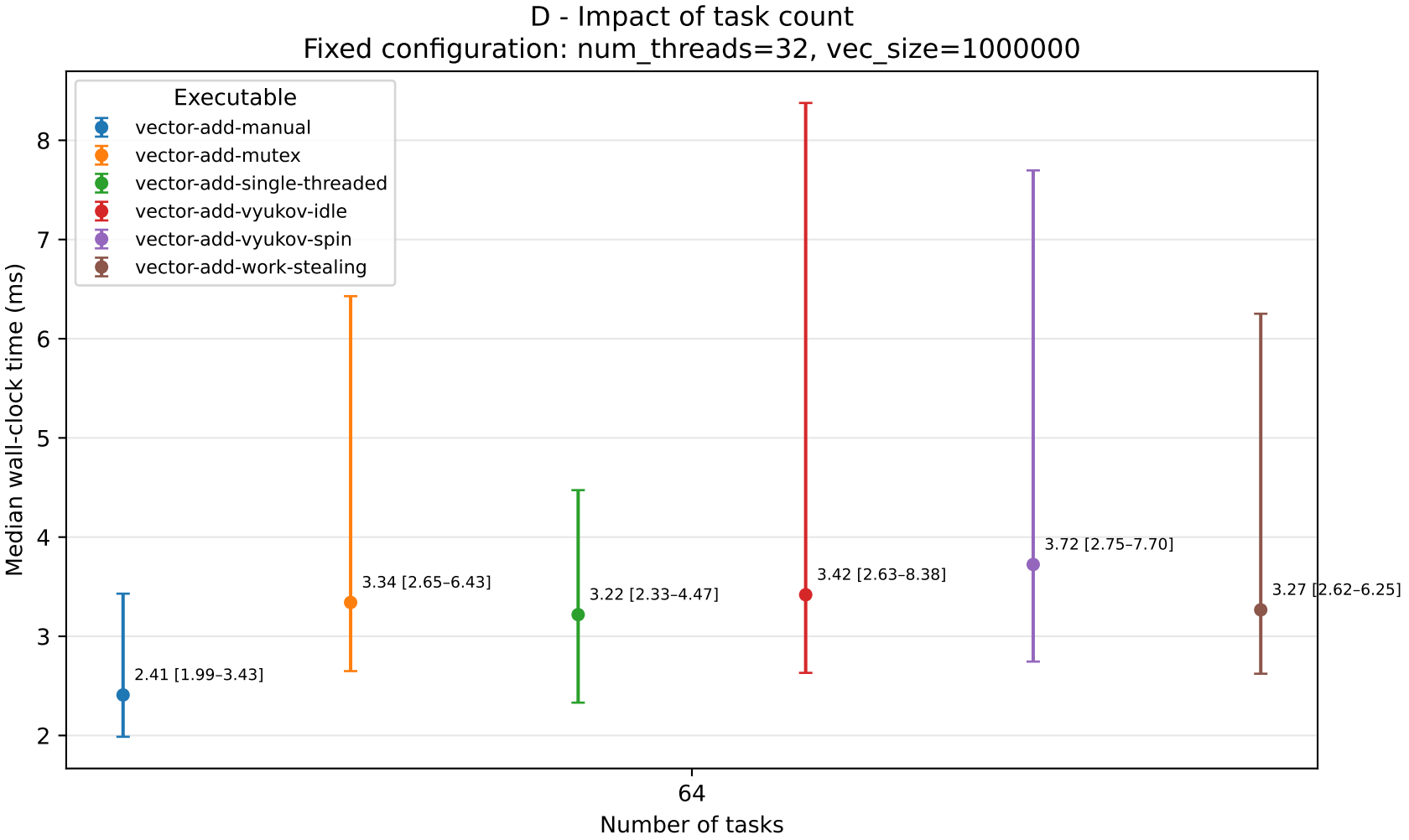

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=100000000
Median wall-clock time (ms)
60.0
57.5
55.0
52.5
50.0
47.5
45.0
42.5
40.0
53.08 [50.01–59.81]
53.15 [50.13–60.18]
41.87 [39.80–51.51]
53.02 [49.11–59.88]
53.06 [48.67–59.87]
53.25 [48.55–59.40]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks

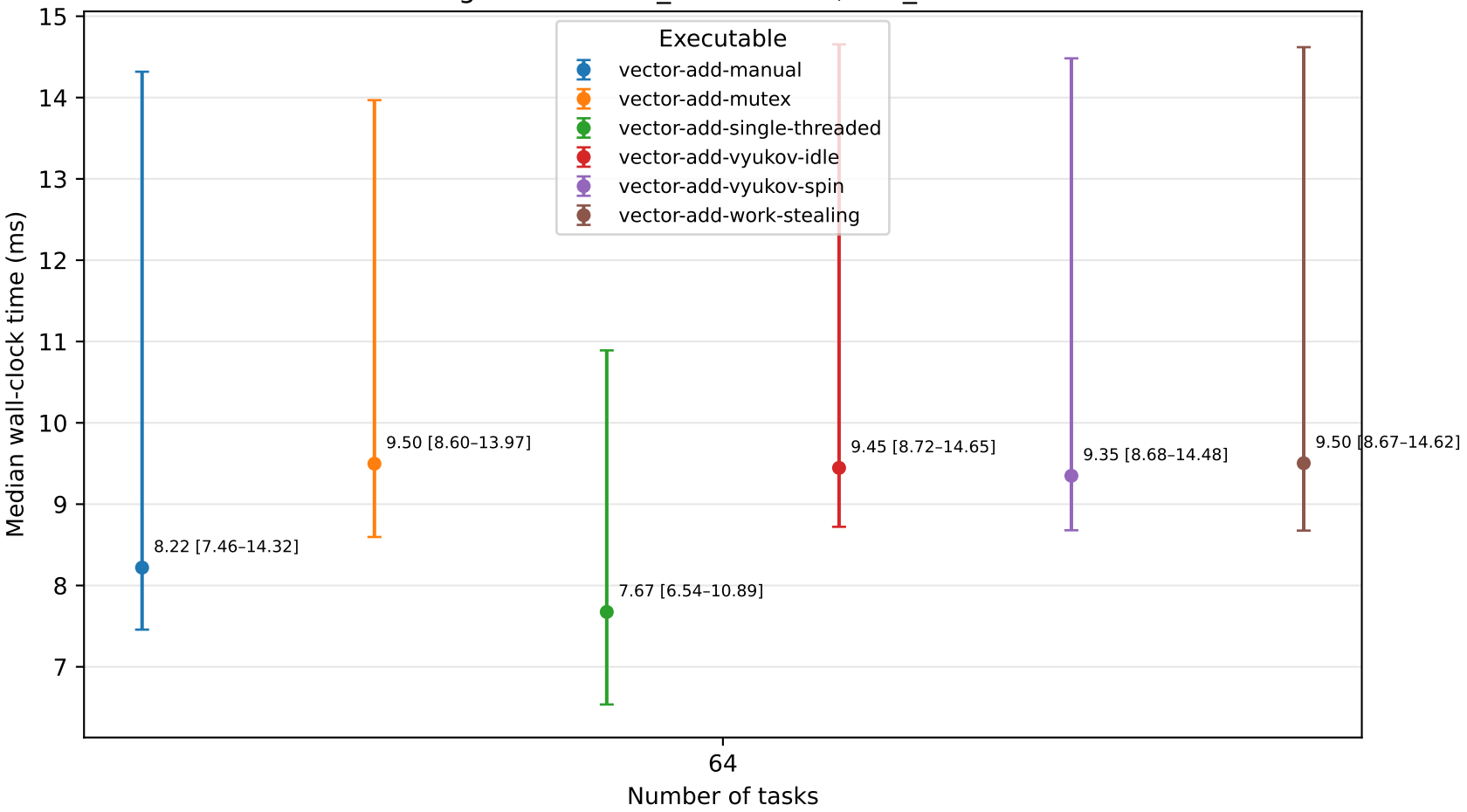

D - Impact of task count
Fixed configuration: num_threads=16, vec_size=200000000
Median wall-clock time (ms)
110
105
100
95
90
85
80
103.91 [97.26–110.01]
102.63 [93.86–109.64]
82.31 [76.98–92.01]
102.55 [93.10–110.99]
102.31 [92.76–111.42]
102.83 [94.59–109.75]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks

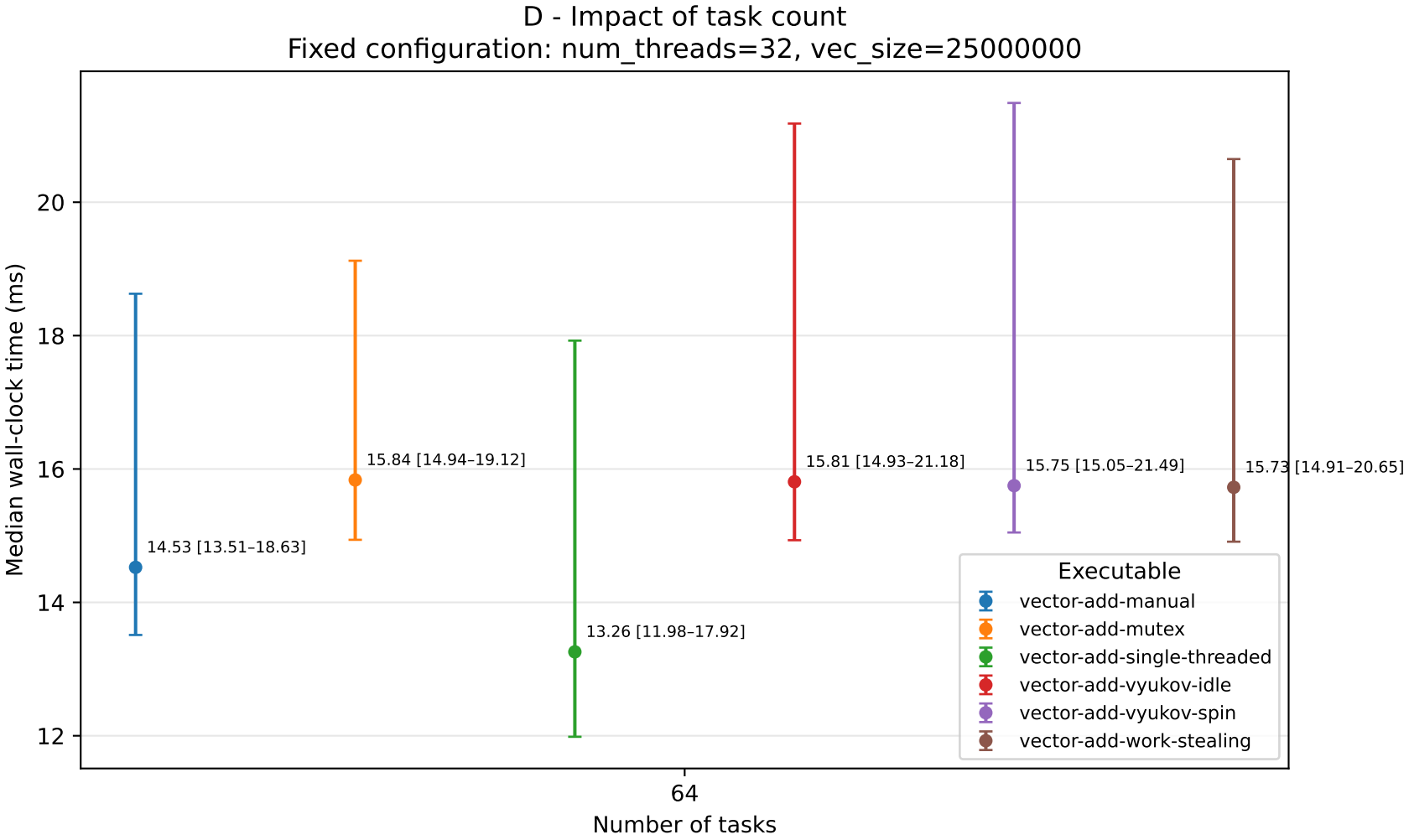

D - Impact of task count
Fixed configuration: num_threads=32, vec_size=992
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
1
2
3
4
5
6
1.79 [1.39–3.07]
2.14 [1.74–4.08]
1.27 [0.70–2.41]
2.32 [1.78–4.97]
2.36 [1.79–6.19]
2.24 [1.65–4.59]
64
Number of tasks

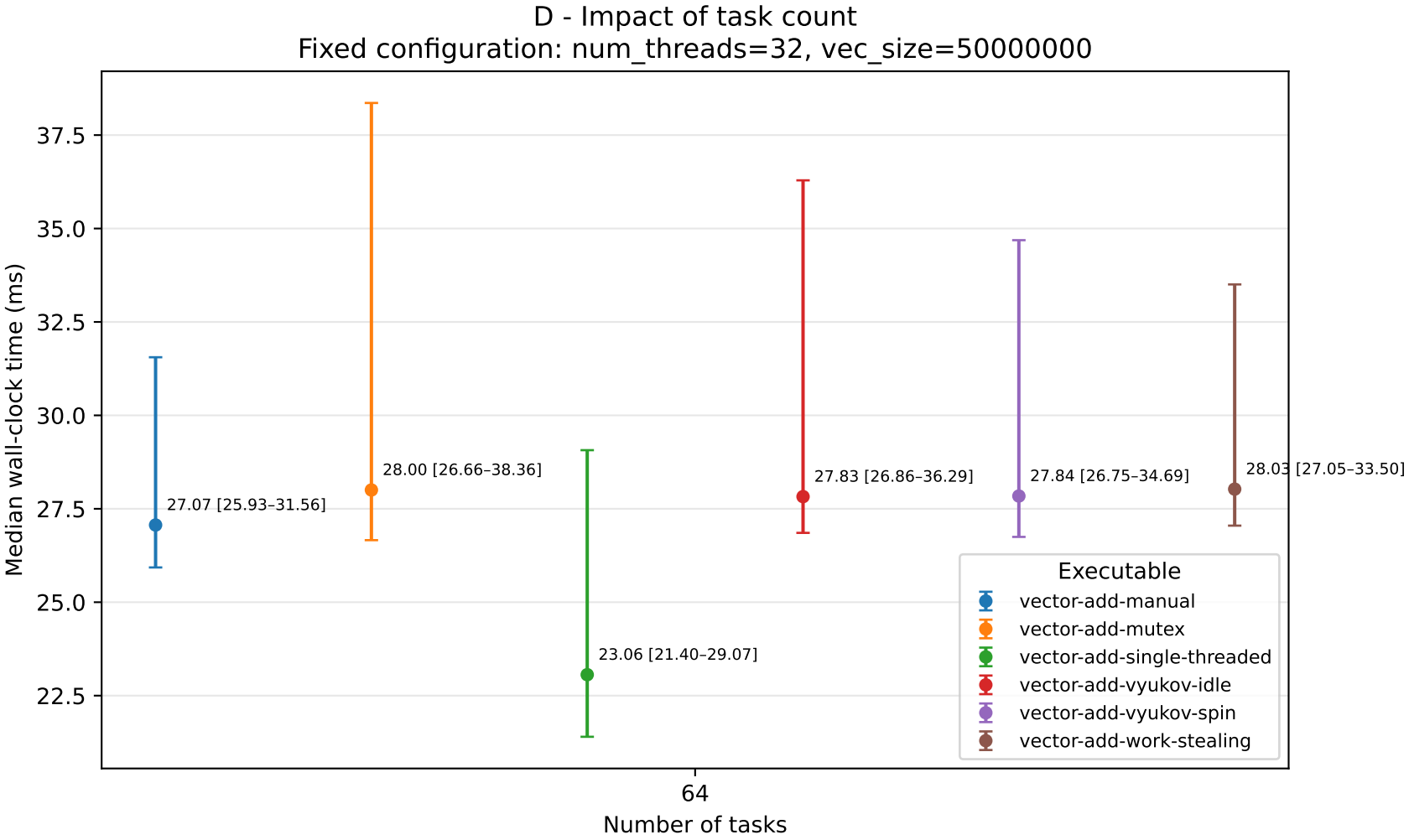

D - Impact of task count
Fixed configuration: num_threads=32, vec_size=10016
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
1
2
3
4
5
1.82 [1.40–3.71]
2.15 [1.75–5.47]
1.17 [0.75–2.70]
2.28 [1.67–4.63]
2.20 [1.79–4.97]
2.15 [1.67–4.79]
64
Number of tasks

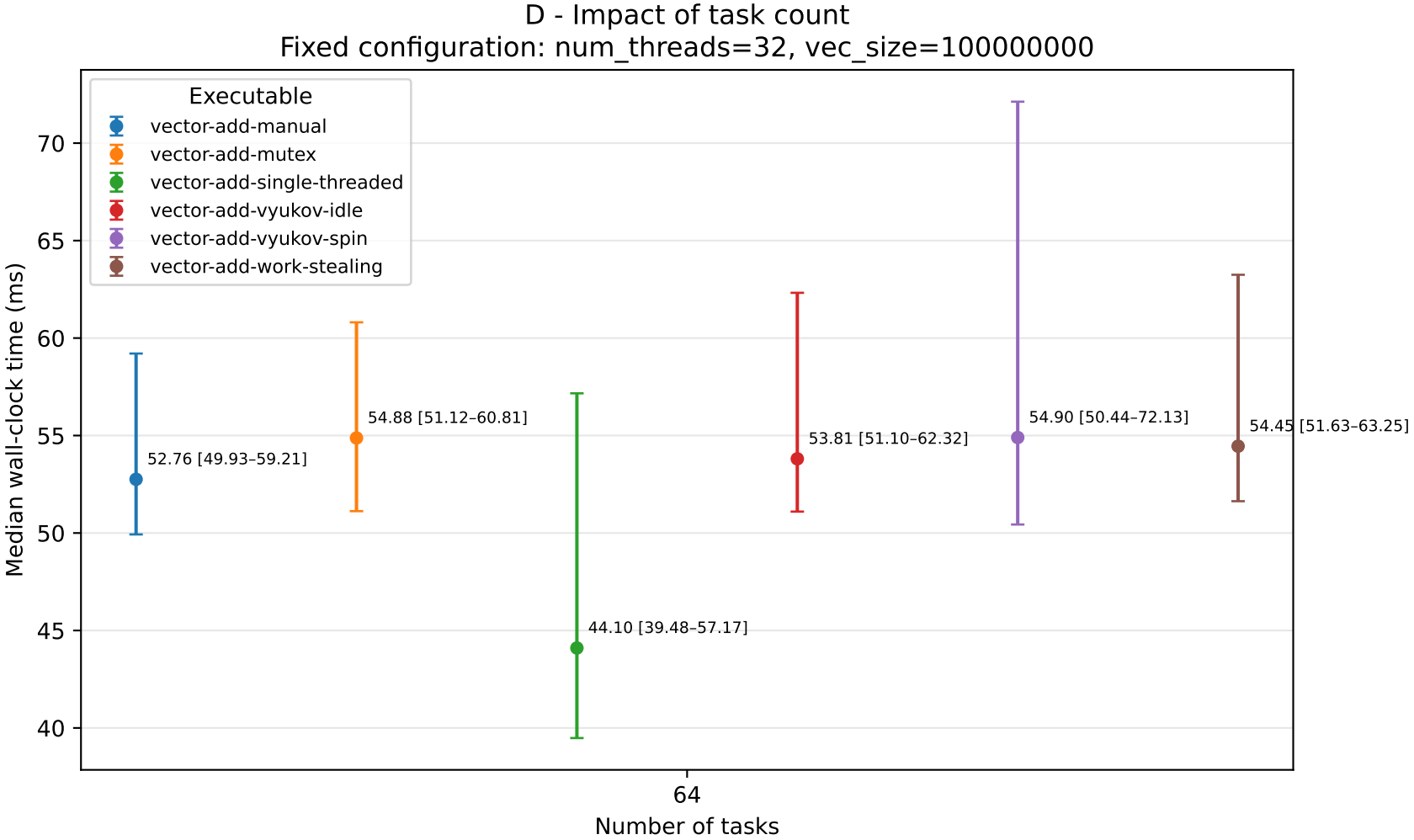

D - Impact of task count
Fixed configuration: num_threads=32, vec_size=100000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
1
2
3
4
5
6
1.83 [1.46–2.74]
2.35 [1.85–3.90]
1.56 [1.03–3.64]
2.98 [1.86–5.93]
2.74 [1.75–5.77]
2.69 [1.85–5.02]
64
Number of tasks

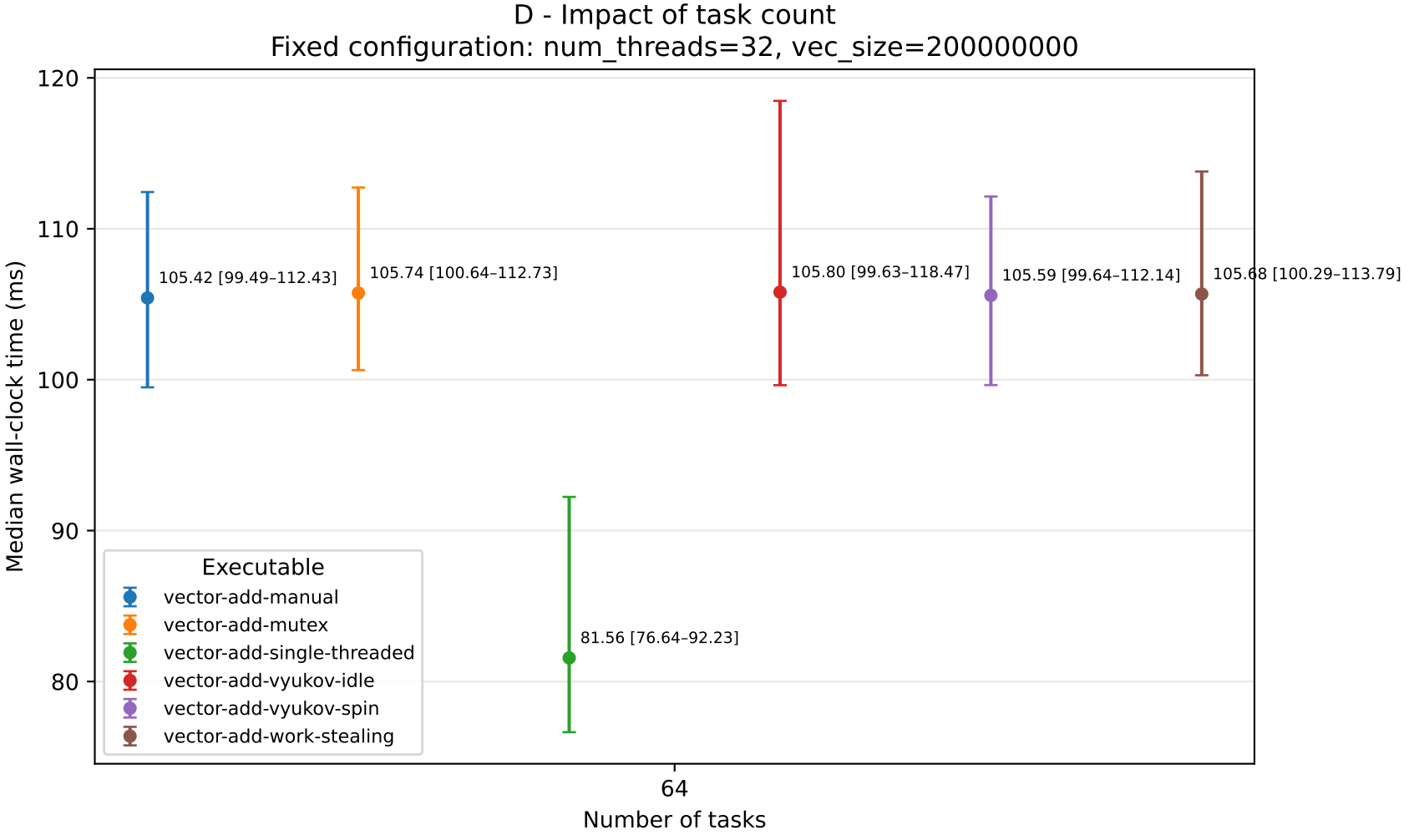

D - Impact of task count
Fixed configuration: num_threads=32, vec_size=1000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
2
3
4
5
6
7
8
2.41 [1.99–3.43]
3.34 [2.65–6.43]
3.22 [2.33–4.47]
3.42 [2.63–8.38]
3.72 [2.75–7.70]
3.27 [2.62–6.25]
64
Number of tasks


D - Impact of task count
Fixed configuration: num_threads=32, vec_size=10000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
15
14
13
12
11
10
9
8
7
8.22 [7.46–14.32]
9.50 [8.60–13.97]
7.67 [6.54–10.89]
9.45 [8.72–14.65]
9.35 [8.68–14.48]
9.50 [8.67–14.62]
64
Number of tasks


D - Impact of task count
Fixed configuration: num_threads=32, vec_size=25000000
Median wall-clock time (ms)
20
18
16
14
12
14.53 [13.51–18.63]
15.84 [14.94–19.12]
13.26 [11.98–17.92]
15.81 [14.93–21.18]
15.75 [15.05–21.49]
15.73 [14.91–20.65]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks


D - Impact of task count
Fixed configuration: num_threads=32, vec_size=50000000
Median wall-clock time (ms)
37.5
35.0
32.5
30.0
27.5
25.0
22.5
27.07 [25.93–31.56]
28.00 [26.66–38.36]
23.06 [21.40–29.07]
27.83 [26.86–36.29]
27.84 [26.75–34.69]
28.03 [27.05–33.50]
64
Number of tasks
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing


D - Impact of task count
Fixed configuration: num_threads=32, vec_size=100000000
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
Median wall-clock time (ms)
70
65
60
55
50
45
40
52.76 [49.93–59.21]
54.88 [51.12–60.81]
44.10 [39.48–57.17]
53.81 [51.10–62.32]
54.90 [50.44–72.13]
54.45 [51.63–63.25]
64
Number of tasks


D - Impact of task count
Fixed configuration: num_threads=32, vec_size=200000000
Median wall-clock time (ms)
120
110
100
90
80
105.42 [99.49–112.43]
105.74 [100.64–112.73]
81.56 [76.64–92.23]
105.80 [99.63–118.47]
105.59 [99.64–112.14]
105.68 [100.29–113.79]
Executable
vector-add-manual
vector-add-mutex
vector-add-single-threaded
vector-add-vyukov-idle
vector-add-vyukov-spin
vector-add-work-stealing
64
Number of tasks